Environmental recurrence selects a costly memory-use policy: long-run population effects and immediate retrieval effects

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Abstract

Prior work establishes that environmental predictability can favor memory and that memory can carry costs. We distinguish selection on a supplied private-cache-use policy, long-run population performance, and immediate effects of retrieval in an abstract computational population. Three prospective studies span two operator bundles. In Study 001, 41,600 paired blocks showed that lag-two recurrence enriched a mutable memory-use allele despite retrieval displacing one of two fresh proposals: its late-window frequency exceeded the neutral 1/2 by 0.310292 (95% familywise interval [0.302482, 0.318102]) and its nonrecurrent-law frequency by 0.441635 ([0.426015, 0.457256]). The recurrent ACTIVE-minus-SHAM accuracy contrast was 0.009377 ([-0.005138, 0.023892]), bounded below one correct bit (1/32), with sign unresolved. Study 002 used a new 41,600-block cohort and a parent-distance-matched control that scrambled cache direction. Directional information increased memory-use frequency by 0.184203 ([0.169688, 0.198718]), while its total accuracy effect was 0.008650 ([-0.006521, 0.023821]), again bounded below one bit with sign unresolved. In Study 003, an independently implemented torus/tournament bundle and a new 41,600-block cohort reproduced recurrence-attributable enrichment, while its recurrent performance contrast, 0.005057 ([-0.009458, 0.019571]), remained below its positive 1/32 maximal-loss scale with sign unresolved. A subsequent prospective BIT/PL experiment, BIT002, used 42,000 new histories and identical-state, paired-tape retrieval interventions at late checkpoints. At recurrent targets, cached rather than fresh retrieval increased eligible-lineage survivor share by 5.041741 percentage points ([3.480912, 6.602570]) and a hashed-selected focal family's unconditional population-normalized contribution by 1.239360 points ([1.001549, 1.477172]), both above their prespecified material thresholds. Immediate population accuracy increased by 2.484087 points ([0.923257, 4.044916]), with magnitude relative to the material and negligibility thresholds unresolved. All three innovation-target effects were bounded within prespecified margins; their negative point estimates and zero-containing intervals do not establish absence. BIT002 intervals have joint coverage at least 95% under the independent-uniform-Philox-word working idealization. Immediate retrieval effects and long-run arm contrasts answer different questions. The findings establish neither long-run mediation of allele enrichment nor an effect of intervening on recurrence, generality across operator bundles, or biological memory.

1. Introduction

Recurring environments make the past informative about the future, and a substantial literature shows that this can favor carrying information across time. Kuijper and Johnstone found that environmental stability and predictability can favor positive parental effects and long-term phenotypic memory under genetically determined transmission rules [1]. Carja and Plotkin showed that, in periodically fluctuating environments, an intermediate degree of phenotypic memory maximizes the fixation probability of an allele that permits heritable phenotypic heterogeneity [2]. Dey, Proulx and Teotónio showed experimentally that temporal environmental structure can select maternal effects [8]; Skanata and Kussell derived evolutionary regimes for response networks with phenotypic memory in random environments [9]; and Kronholm found that anticipatory effects mediated by epigenetic changes evolve when an environmental cue is reliable, the environment changes at intermediate rates and the costs of the effect are low [7].

Models of learning reach related conclusions. Wright and colleagues allowed memory weighting and current sampling effort to evolve under explicit costs and found that memory increased with temporal autocorrelation and interacted with costly sampling [3]. Arehart and Adler quantified the costs and benefits of memory duration and exploration in a minimal learning model and found that the marginal benefit of learning is often small [4]. Morgan, Suchow and Griffiths inserted human participants as agents into evolutionary simulations with simulated loci for learning and memory and found that rapid environmental change dampened their evolution [5]. Among the studies identified by our bounded search, this is a close precedent for an evolving memory-use trait, although its learners were human participants rather than a cleanly mutable allele. Other work is at least as close on other dimensions: Carja and Plotkin analyzed a mutable modifier allele and its fixation in a finite population [2], and Grabowski and colleagues studied the early evolution of memory usage in digital organisms [6].

The value and cost of memory that is already present are also well studied. Dunlap and Stephens linked environmental change to optimal sampling and memory weighting [10]. Lambert and Kussell measured bacterial memory under fluctuating carbon sources and modeled its long-term fitness benefit [11]. Rescan and colleagues linked phenotypic memory to population growth and extinction risk in a noisy environment [12]. George optimized memory-driven phenotypic adaptation in fluctuating environments [13], and Jain, Jolly and George added adaptation costs [14]. Pollack, Nozoe and Kussell analyzed selection on gene regulation and phenotypic memory under metabolic fluctuations [15]. Leadbeater and Watrobska emphasize that memory can be shaped by selection along many mechanistic dimensions rather than as a single scalar trait [16].

Against this background, we do not claim to show that memory evolves, that predictability favors memory, or that costly memory can be selected; those propositions are occupied by the work above. The contribution here is a specific joint design, executed with a new implementation and new random cohorts, that allows two outcomes to be judged separately. First, the memory architecture, a one-record private cache, is supplied, and only a binary use policy mutates and is selected, in a finite population with explicitly stochastic survival. Second, the cost of using memory is a one-for-one search-budget opportunity cost: a carrier that retrieves its cache gives up one of its two fresh proposals, with no added fitness penalty or tunable coefficient. Third, a label-blind SHAM arm supplies an exact neutral benchmark for the label process at the same query budget. Fourth, allele enrichment and population accuracy were judged by prospectively separate decision rules, each with a stated meaningful scale. Fifth, a recurrence-gated, parent-distance-matched control separates the directional content of a retrieved record from the loss of the fresh proposal it displaces. Sixth, the central Study 001 classification was tested again, prospectively, in an independently implemented operator bundle with a different alphabet, loss,

proposal rule and survival rule and with entirely new cohorts.

The separation matters because the two outcomes need not move together. An allele can rise in frequency because it improves its carriers' chances of being among the survivors while the population's average fit to the environment changes little. Carriers may displace non-carriers that would have achieved similar accuracy, or gains at recurrent updates may be offset by lost exploration at non-recurrent updates. Treating allele enrichment as evidence of population benefit would conflate these possibilities.

We first report the accepted results of three prospective studies in two operator bundles. In Study 001 (PHASE2-MUTABLE-MEMORY-001), under one recurrent target law that was deliberately aligned with the cache delay, lag-two recurrence strongly enriched the mutation-generated memory-use allele above both its exact neutral expectation and its frequency under a nonrecurrent control, despite the fixed opportunity cost. The corresponding ACTIVE minus SHAM accuracy contrast was bounded below the prespecified one-correct-bit scale, with sign unresolved, and the recurrence interaction in accuracy remained unresolved. Study 001 could not say how much of that selection responded to what the cache contained rather than to the fresh proposal that retrieval displaced. Study 002 (PHASE2-PERFORMANCE-CONVERSION-002), a post-v1.0 prospective extension with a new random cohort, addressed that question within the aligned law by comparing directional retrieval (INFO) with a recurrence-gated control that keeps each retrieved candidate at the cache's distance from its parent but scrambles its direction (NONINFO). Directional information strongly increased selection on the memory-use policy, yet its total effect on population accuracy was bounded below the one-bit scale with sign unresolved. Studies 001 and 002 share one operator bundle, so neither could say whether the separation depends on its binary alphabet, Hamming loss, XOR-mask proposals and weighted survival. Study 003 (PHASE2-TORUS-MEMORY-003), a post-v1.1 prospective extension, re-tested the Study 001 primary classification in one alternative bundle, a 32-locus torus with 2^16 alleles per locus, graded circular loss, uniform fresh proposals and four-entry rank-tournament survival, implemented anew with new random addressing and cohorts. Recurrence-attributable enrichment recurred, and the recurrent-law performance contrast was again bounded below the positive 1/32 scale with sign unresolved. Because alphabet, loss, resampling and survival changed together, Study 003 cannot attribute this outcome to any one of them. We present the three studies as a single result: in the first operator bundle, recurrence selected a costly memory-use policy, directional cache information strongly amplified that selection, and neither was accompanied by a demonstrated commensurate population-performance advantage at the fixed scale; in one independently implemented alternative bundle, recurrence-attributable enrichment and a bounded performance contrast recurred together.

A subsequent prospective experiment, PHASE3-BIT-RETRIEVAL-CAUSALITY-002 (BIT002), asks a different question: at the same late BIT/PL state and the same operator tape, what changes in the next turnover when valid cached probes are replaced with paired fresh probes? It adds direct retrieval interventions while retaining the accepted long-run findings as separate evidence. BIT002 is named by its study identifier throughout; it is not a second name for Study 002 (PHASE2-PERFORMANCE-CONVERSION-002), an additional operator bundle, or completion of the original broader Phase 3 protocol. Its checkpoints, cohorts, estimands and interval family were frozen before production; the decision to integrate its accepted results into this manuscript was made afterward.

2. Model and prospective design

This paper reports three prospective studies in two operator bundles. Sections 2.1-2.7 describe Study 001, Section 2.8 describes the Study 002 extension, which changes only the construction of one candidate, and Section 2.9 describes Study 003, which poses the Study 001 question again in an independently implemented alternative operator bundle. For each study, the specification, decision table, cohort size, audit plan and claim limits were frozen before any production outcome was generated. For Study 001, a non-implementing AI reviewer returned a terminal GO review of the frozen specification before production was authorized; the specification hash, review record and timestamps are given in Section 7. In no study was any parameter, cohort, horizon, target law, mutation rate or cost changed after outcomes were observed. The complete update order is given in Supplementary Sections S2-S7 for Study 001, S17-S18 for Study 002 and S28-S31 for Study 003.

2.1 Population, search state and objective

The population contains 32 haploid individuals in fixed slots 0-31. Each individual carries an unsigned 32-bit genotype and a binary policy allele, M (use the private cache) or F (fresh exploration). An M individual also holds a private cache that is either invalid or contains one 32-bit genotype; F individuals hold no valid cache. The allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval. Lineage identifiers are audit metadata only and are never read by any scientific operator. Each path runs for updates 1-256. At update t the environment presents a 32-bit target T_t, and the raw objective of a genotype is its Hamming mismatch to T_t. Population accuracy at an update is one minus the mean survivor mismatch divided by 32, measured after survival; a population of uniformly random genotypes has expected accuracy 1/2. The late window is updates 193-256 inclusive.

The population size, bit length, four-candidate architecture, lag-two law and selection weights were inherited from earlier internal private-memory studies so that the changed question, selection on a mutable use policy, would be identifiable. The implementation and all random cohorts are new and import no producer code from those studies. This study is not a replication of them.

2.2 Candidate budget and the one-proposal opportunity cost

Every parent produces exactly four evaluated candidates per update, so every update makes exactly 128 objective queries regardless of allele, arm or law. The candidates are (1) the unchanged parent; (2) a policy probe; (3) a global scout, the parent XOR an independent uniform 32-bit mask; and (4) a local child, made by taking the lowest-mismatch member of candidates 1-3 (ties broken by a keyed random key and then by family order) and flipping each of its 32 bits independently with probability 1/32. For an F individual the policy probe is the parent XOR a second independent uniform mask, so an F carrier makes two fresh, effectively uniform proposals per update. For a valid M carrier in the ACTIVE arm, the policy probe is the cached genotype itself. Because the probe is also eligible to be the donor of the local child, a useful cache can shape two of the four candidates.

The cost of memory is therefore an opportunity cost on a fixed search budget: retrieval replaces exactly one of two effectively uniform fresh proposals with the cached record. There is no score penalty and no tunable cost coefficient. This is a model-specific search-budget trade-off, not a calibrated energetic, metabolic or biological cost, and its magnitude is conditional on the frozen

fresh-probe distribution. When the cache is uninformative, an M carrier loses one fresh draw at the same query count. When the cache closely matches the current target, retrieval supplies a candidate that a uniform fresh draw is unlikely to match. Genotypes are not deduplicated: if the cache equals the unchanged parent, both copies are separately evaluated and the update still makes 128 queries.

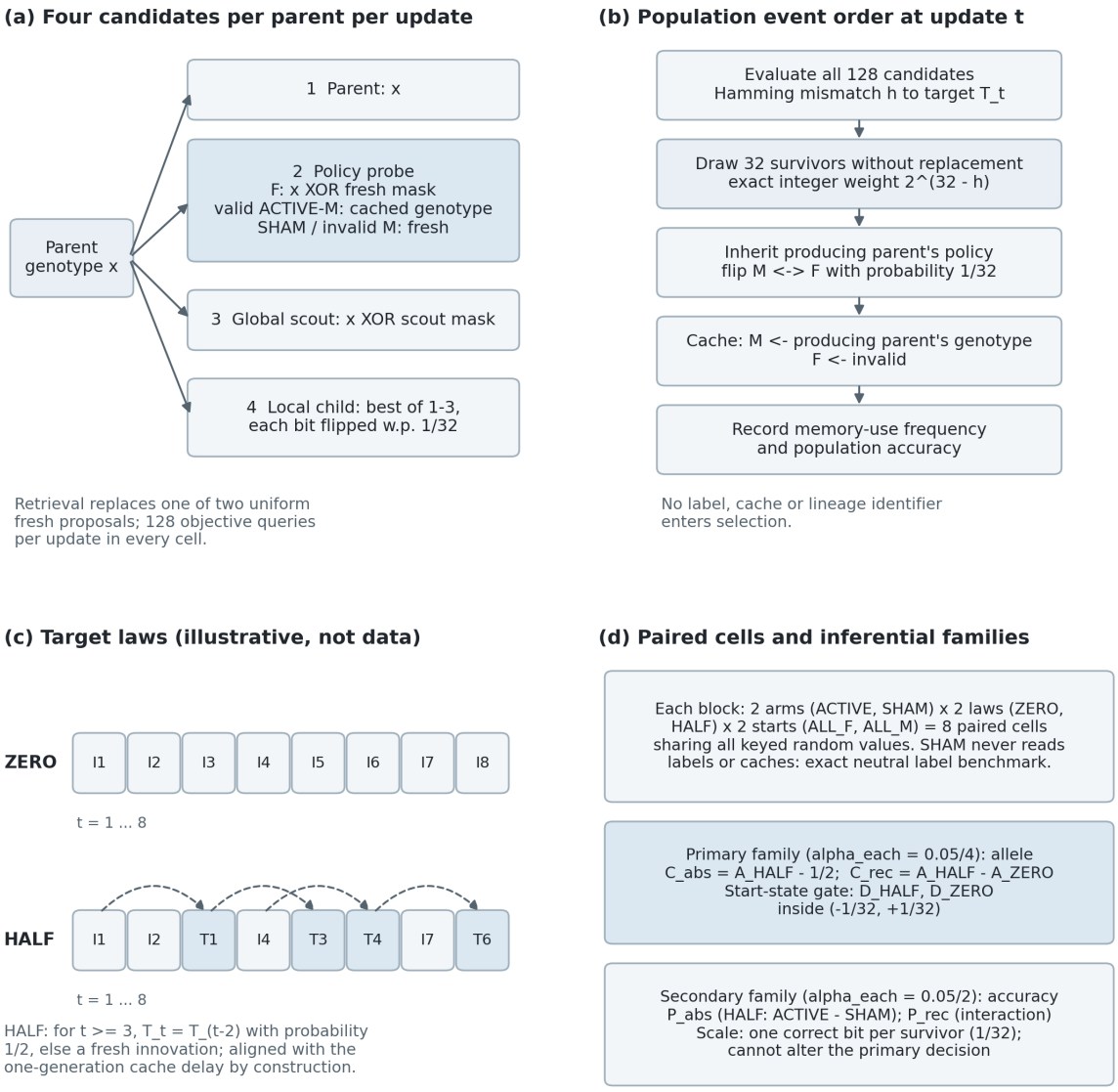


Figure 1. Model and prospective design. (a) One update for one parent: four evaluated candidates, of which only the policy probe depends on the allele and arm; a valid ACTIVE-M carrier replaces one of its two fresh uniform proposals with its cached genotype. (b) Population-level event order: evaluation of 128 candidates against the current target, exact integer-weighted sequential sampling of 32 survivors, symmetric policy mutation, and cache assignment. (c) The two target laws; HALF returns to the target from two updates earlier with probability 1/2, which aligns with the one-generation cache delay by construction. (d) The eight paired cells per block and the separate primary (allele) and secondary (accuracy) inferential families. The schematic contains no outcome values. Panels (a), (b) and (d) depict the operator bundle shared by Studies 001 and 002; the Study 003 bundle is described in Section 2.9 and compared element by element in Supplementary Section S27.

2.3 Target laws and the deliberate HALF alignment

For every block and update, an innovation I_t (a uniform 32-bit word) and a fair recurrence bit R_t are generated whether or not they are used. Under ZERO, T_t = I_t at every update. Under HALF, T_1 = I_1 and T_2 = I_2; for t >= 3 the target returns to T_(t-2) when R_t = 1 and otherwise equals I_t. Non-recurrent HALF updates use the same innovation as ZERO at that update.

The cache of a survivor records its producing parent's genotype from the update in which the survivor was made, and that parent had itself survived selection against the previous target. A cache used at update t+1 therefore holds a genotype that survived selection against T_(t-1), which under HALF is the target at update t+1 with probability 1/2. HALF was a prior, outcome-informed choice made precisely to align the one-generation cache delay with a two-update target return; it favors retrieval by construction. No other lag, copy probability, target law, cost or model class was tested.

2.4 Selection, policy mutation and cache inheritance

All 128 candidates are evaluated. Candidate j with mismatch h_j receives the exact integer weight 2^(32-h_j), proportional to 2^(-h_j). Thirty-two survivors are drawn sequentially without replacement, each with probability proportional to its weight among the candidates not yet selected. This is an exact Plackett-Luce weighted sample implemented entirely in unsigned integer arithmetic, with unbiased bounded integers produced by multiply-high rejection (Supplementary Section S5). No label, cache, lineage identifier or penalty enters selection; the policy allele affects survival only through the genotypes it causes to be proposed.

Each survivor first inherits its producing parent's policy. Then, independently in each survivor slot, the policy flips M <-> F with probability mu = 1/32. After mutation, an F survivor has an invalid cache and an M survivor's cache becomes its producing parent's pre-update genotype. From update 2 onward every M carrier therefore holds a valid one-step lineage cache. A new F -> M mutant first uses its record on the next update, and an M -> F -> M history carries no older hidden record. Allele frequencies and accuracy are recorded after this transition.

2.5 Paired cells and the exact SHAM benchmark

Each of 41,600 independent blocks contains the same eight paired cells: information arm (ACTIVE or SHAM) x target law (ZERO or HALF) x policy start (ALL_F or ALL_M). Within a block, all eight cells share 32 initial uniform genotypes and all coordinate-keyed random values; caches start invalid. Random values come from counter-based Philox4x32-10 streams indexed by purpose, block, update, entity and subindex, and every draw is generated whether or not it is used, so no label, branch, arm or law changes any later random value (Supplementary Section S7).

In SHAM, the policy probe is always the fresh construction, so genotypes never depend on a label or cache. Two exact identities follow. N1: SHAM genotype, mismatch, accuracy and survivor trajectories are bit-identical between the ALL_F and ALL_M starts. N2: SHAM labels in the two starts are exact complements after every update, so their memory-use frequencies sum to 1 at every update and their start-averaged late-window frequency is exactly 1/2 in every block. With symmetric mutation and r = 1 - 2mu = 15/16, the SHAM expectations are P(M at t | ALL_F) = [1 - r^t]/2 and P(M at t | ALL_M) = [1 + r^t]/2. The expected ALL_M minus ALL_F difference is r^256 = 6.678005283720385e-08 at update 256 and r^193(1 - r^64)/(64(1 - r)) = 9.57983820600146e-07 averaged over the late window.

SHAM is an exact neutral benchmark for the label process (N1, N2; expected start-averaged frequency 1/2) and a paired control for query budget and keyed random draws. Because ACTIVE

genotypes and genealogies diverge from SHAM at the first valid-M retrieval, SHAM does not hold realised drift or genotype state fixed for ACTIVE paths; ACTIVE-SHAM contrasts are arm-level comparisons within the model, not path-level counterfactuals. It does not by itself separate the information carried by the cache from the mere loss of one fresh proposal; the ZERO law, under which the cache carries no lag-two information, supplies that contrast through C_rec, and Study 002 (Section 2.8) supplies a within-law contrast. It does not represent any alternative cost, retrieval rule or memory architecture. Its neutral formulas justify the late window in expectation for SHAM only and do not establish that ACTIVE paths have converged.

2.6 Estimands, intervals and frozen decisions

For each block, let L(arm, law, start) be the mean memory-use frequency over updates 193-256. Define A_HALF = [L(ACTIVE, HALF, ALL_F) + L(ACTIVE, HALF, ALL_M)]/2 and A_ZERO analogously. The four primary estimands are:

- C_abs = A_HALF - 1/2, the recurrent-law excess over the exact neutral expectation;
- C_rec = A_HALF - A_ZERO, the excess attributable to recurrence;
- D_HALF = L(ACTIVE, HALF, ALL_M) - L(ACTIVE, HALF, ALL_F), the start-state contrast under HALF; and
- D_ZERO = L(ACTIVE, ZERO, ALL_M) - L(ACTIVE, ZERO, ALL_F), the start-state contrast under ZERO.

Let P(arm, law) be start-averaged late-window mean population accuracy. The two secondary estimands are:

- P_abs = P(ACTIVE, HALF) - P(SHAM, HALF), the recurrent-law accuracy contrast; and
- P_rec = [P(ACTIVE, HALF) - P(SHAM, HALF)] - [P(ACTIVE, ZERO) - P(SHAM, ZERO)], the recurrence interaction in accuracy.

The meaningful allele scale is Delta = 1/32 = 0.03125, one expected individual in the 32-member population; it refers to a late-window expected excess, not to the persistence of one named carrier. The meaningful performance scale is Delta_P = 1/32, one additional correct bit per survivor on average, which is 3.125 percentage points of accuracy.

Each family is controlled at 95% familywise coverage by two-sided Hoeffding intervals with a Bonferroni split. For a block variable with range length R and n = 41,600 independent blocks, the half-width is h = R sqrt[ln(2/alpha_each)/(2n)]. In the primary family, alpha_each = 0.0125; C_abs has range length 1 and half-width 0.00781023, and C_rec, D_HALF and D_ZERO have range length 2 and half-width 0.01562046, just below Delta/2 = 0.015625. In the secondary family, alpha_each = 0.025; P_abs has range length 2 and half-width 0.01451463, and P_rec has range length 4 and half-width 0.02902925. No empirical variance, normal approximation, within-path independence, pilot or favorable pairing enters these bounds. The two families are controlled separately, not jointly.

The primary decision table was applied in order: (1) failure of any deterministic identity, seed or counter audit, saved-output reconstruction or query count gives INVALID; (2) if either D_HALF or D_ZERO has an interval not wholly inside (-Delta, +Delta), the result is START-DEPENDENT and the question unresolved; (3) if the lower bounds of both C_abs and C_rec exceed Delta, the result SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT; (4) if the lower bound of C_abs exceeds Delta and the upper bound of C_rec is at or below Delta, enrichment is not attributable to recurrence at the fixed scale; (5) if the upper bound of C_abs is at or below Delta, the result is a bounded negative at the one-individual scale; (6) every other pattern is unresolved. Each

secondary estimand was classified, in order, as meaningful positive performance if its lower bound exceeds Delta_P, meaningful adverse performance if its upper bound is below -Delta_P, BOUNDED BELOW THE POSITIVE ONE-BIT SCALE if its upper bound is at or below Delta_P, and UNRESOLVED otherwise. Secondary classifications cannot alter the primary decision.

Sixteen absolute cell means, the late-window memory-use frequency and accuracy for each of the eight cells, were saved as descriptive records without inferential bounds. Together with the six inferential records they form the 22 prespecified saved estimate records. The descriptive means support interpretation only.

2.7 The start-state gate, and why allele frequency is not accuracy

The start-state gate, rule 2, asks whether populations started entirely from F and entirely from M reach the same late-window ACTIVE frequency to within one expected individual. If they did not, the late window would partly reflect where paths began rather than a late-window state approached from both starts within the frozen horizon, and no enrichment classification would be made. The gate is applied before any enrichment rule. Passing it shows only that both late-window start contrasts were bounded within (-1/32, +1/32) at 95% familywise coverage over the 256-update horizon; it does not demonstrate start-independence in general, stationarity or a unique long-run state.

Allele frequency measures which policy the survivors carry; accuracy measures how well the survivors match the target. The two are linked only through the genotypes that M carriers propose. A policy can raise its carriers' representation among survivors by supplying high-scoring candidates at recurrent updates while costing a fresh proposal at every update, so its effect on mean survivor accuracy can be smaller than, or differently signed from, its effect on its own frequency. This is why the two outcomes were assigned separate estimands, scales and decision rules in advance.

2.8 Study 002: separating directional cache information from displacement

Study 002 retains every Study 001 operator: the population, genotype, mutation rate 1/32, cache inheritance, four-candidate budget with 128 objective queries per update, exact integer-weighted survival, 256-update horizon and late window. It uses only the HALF law, which remains deliberately aligned to the cache delay; there is no ZERO arm. It changes only the policy probe, in three arms. For a parent with genotype x and valid cache c:

- **INFO:** a valid M carrier proposes c, exactly as in the Study 001 ACTIVE arm;
- **NONINFO:** a valid M carrier proposes c at innovation updates (t < 3 or R_t = 0), and at recurrent updates (t >= 3 and R_t = 1) proposes x XOR pi_(b,t)(c XOR x), the cache displacement with its bit positions moved by a shared permutation; and
- **SHAM:** every label uses the fresh construction, as in Study 001.

F carriers and invalid-cache M carriers use the fresh construction in every arm. The experimental operator reads R_t; the evolving policy cannot. INFO and NONINFO therefore follow identical rules at innovation updates and differ only in whether a valid cache displacement keeps its direction at recurrent updates. Both pay the same one-proposal displacement cost at every valid-M use.

pi_(b,t) is one exact uniform random permutation of the 32 bit positions, generated for every block b and update t regardless of branch or arm by a Fisher-Yates shuffle with unbiased multiply-high rejection, and shared by all cells and all parents of that block and update (Supplementary Section S18). Because permuting bit positions preserves Hamming weight, a NONINFO decoy lies exactly as far from its parent as the cache does. The shared permutation also

preserves overlaps among the displacement masks of different parents under one common relabeling of coordinates, but it does not preserve every cross-genotype relation; for example, siblings that share one cached genotype do not share one decoy endpoint. A displacement of weight 0 or 32 cannot be scrambled and retains its direction; these cases are counted in the diagnostics.

Each of 41,600 new independent blocks contains six paired cells, arm (INFO, NONINFO, SHAM) x policy start (ALL_F, ALL_M), sharing 32 initial uniform genotypes and every keyed random value at compatible coordinates, including the permutations. Random values come from the same Philox4x32-10 construction under a new namespace; no genotype, target, trajectory, block, random tape or outcome from Study 001 was reused. The Study 002 producer extends the authenticated Study 001 producer, and its unchanged operators were required to remain fixture-identical. SHAM obeys the same identities N1 and N2 as in Study 001, so its start-averaged late-window memory-use frequency is exactly 1/2 in every block; it serves both as the neutral allele benchmark and as the no-memory performance reference.

For block b, let P(arm, start) be the late-window mean population accuracy and L(arm, start) the late-window mean memory-use frequency, and let P_arm and A_arm be their equal-weight averages over the two starts. The primary family contains three estimands:

- Delta_P = P_INFO - P_NONINFO, the total population-accuracy effect of directional cache information beyond the parent and matched displacement length, mediated through evolved policy use; and
- D_INFO = L(INFO, ALL_M) - L(INFO, ALL_F) and D_NONINFO = L(NONINFO, ALL_M) - L(NONINFO, ALL_F), the finite-horizon start-state gate.

INFO and NONINFO share policy rules but not realized policy states, so Delta_P includes any divergence in policy frequency, cache use, survivor composition and descendants. It is a total effect, not a per-use value-of-information estimate. Two secondary families cannot alter the primary decision. The allele family contains E_INFO = A_INFO - A_NONINFO, the allele-frequency effect of cache direction, and E_NONINFO = A_NONINFO - 1/2, selection under the matched control relative to exact SHAM neutrality. The no-memory performance family contains B_INFO = P_INFO - P_SHAM and B_NONINFO = P_NONINFO - P_SHAM.

The meaningful scale is again 1/32: one correct bit in mean accuracy (3.125 percentage points) or one expected individual in allele frequency. Intervals are two-sided Hoeffding-Bonferroni intervals over n = 41,600 blocks. The primary family uses alpha_each = 0.05/3; each primary block variable has range length 2, giving half-width 0.01517128, below Delta/2 = 0.015625. Each secondary family uses alpha_each = 0.025; E_INFO, B_INFO and B_NONINFO have range length 2 and half-width 0.01451463, and E_NONINFO has range length 1 and half-width 0.00725731. The three Study 002 families are controlled separately from one another and from the Study 001 families.

The frozen primary decision order was: (1) failure of any deterministic identity, coordinate, query-count, saved-output or audit check gives INVALID; (2) if either gate interval is not wholly inside (-1/32, +1/32), the result is START-DEPENDENT and the primary unresolved; (3) a lower bound of Delta_P above 1/32 gives a meaningful positive directional-information effect relative to NONINFO; (4) an upper bound below -1/32 gives a meaningful adverse effect; (5) an upper bound at or below 1/32 gives BOUNDED BELOW THE POSITIVE ONE-BIT SCALE, with a separate statement of whether the interval lies wholly inside (-1/32, +1/32); (6) otherwise UNRESOLVED. Each secondary estimand was classified against the same scale. Even a positive primary would have established a benefit relative to no-memory use only if the lower bound of B_INFO also

exceeded 1/32. No sample extension, parameter change or follow-up cohort follows any branch.

2.9 Study 003: the same question in an alternative operator bundle

Study 003 asks whether the Study 001 classification, recurrence-attributable enrichment of a mutation-generated private-cache-use allele under a one-proposal opportunity cost, recurs in one independently implemented operator bundle that changes the alphabet, the loss, the proposal semantics and the survival rule together. It keeps every element that defines the question: 32 haploid individuals; the supplied one-record lineage cache; the M/F policy allele, which couples retention and retrieval; symmetric policy mutation at rate 1/32; four evaluated candidates per parent and 128 objective queries per update; replacement of exactly one fresh proposal by the cache; the ZERO and HALF target laws, with HALF still deliberately aligned to the cache delay; the 256-update horizon and late window; and the eight paired cells, ACTIVE or SHAM x ZERO or HALF x ALL_F or ALL_M, in each of 41,600 new independent blocks.

Each phenotype is a vector of 32 circular loci, each an unsigned 16-bit integer in Z/2^16 Z. For locus i with u = (x_i - t_i) mod 2^16, the circular distance is d_i = min(u, 2^16 - u), and individual loss is the exact integer Q = sum_i d_i^2, so loss grows gradually with distance on the torus. The policy probe of an F carrier, of an invalid-cache M carrier and of every label in SHAM is an independent uniform 32-coordinate fresh vector; a valid ACTIVE-M carrier proposes its cached phenotype instead. The global scout is a second independent uniform vector. The local child takes the smallest-loss member of the first three candidates and independently replaces each coordinate, with probability 1/32, by a uniform 16-bit value. This is large-alphabet resampling, not a small torus step, and the fresh proposals are uniform rather than local: torus geometry enters through graded loss, not through proposal generation.

Survival uses rank only. For each of the 32 survivor slots, four entries are drawn independently, with replacement, from the 128 candidates; the entry with the lowest exact loss wins, ties being broken by a label-blind keyed 64-bit candidate tie key and then by candidate index. A candidate can win zero, one or several slots. Tournament size four was frozen because 32 slots x four entries equals the 128-candidate pool; it was not calibrated to the survival strength of Study 001. The unique best candidate has expected offspring 8290815/8388608, just below one, and leaves no descendant with probability above one-third, so survival is much weaker than near-deterministic preservation of a clearly best candidate. Each survivor inherits its producing parent's policy, flips M <-> F with probability 1/32 and, if M after mutation, caches its producing parent's pre-update phenotype; there is no phenotype mutation after survival. No label, cache or lineage enters ranking or tie-breaking. Population accuracy is the exact rational (2^40 - total survivor loss)/2^40.

The entire implementation, including canonical Random123 Threefry4x64-20 addressing keyed by purpose with counters (block, update, entity, subindex), was written anew from the frozen specification without importing or adapting producer, analyzer or auditor code from Studies 001 or 002, and every phenotype, target, random tape and cohort is new. SHAM never reads a label or cache when constructing phenotypes, so identities N1 and N2 and the neutral expectations of Section 2.5 hold unchanged.

The estimands C_abs, C_rec, D_HALF, D_ZERO, P_abs and P_rec, their range lengths and the two Hoeffding-Bonferroni families (alpha_each = 0.05/4 and 0.05/2 over n = 41,600 blocks) are those of Section 2.6, so the half-widths are again 0.00781023, 0.01562046, 0.01451463 and 0.02902925. The allele scale is again Delta = 1/32, one expected individual. The performance scale Delta_P = 1/32 is 1/32 of the maximal total loss: on average, moving one coordinate per individual from the torus antipode to exact agreement, or removing roughly three typical

random-coordinate losses per individual. It is not a correct bit, so Study 003 effect magnitudes are not compared numerically with those of Studies 001 or 002; only the prespecified within-bundle classifications are compared. The primary decision order is that of Section 2.6, with rule 3 reading SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT IN THIS ALTERNATIVE OPERATOR BUNDLE; only rule 3 reproduces the Study 001 classification, and a start-dependent or unresolved result would have been no answer rather than evidence against the mechanism. The performance classification order is that of Section 2.6 and cannot alter the primary decision. A crossed interpretation was also frozen: primary support with P_abs bounded below the positive scale would mean that the selection/performance separation recurs in this operator bundle, and primary support with a meaningfully positive P_abs would mean that the separation is bundle-specific at the declared scales. No branch permitted added cohorts, another tournament, another operator class or a component-isolating follow-up. Supplementary Section S27 compares the two operator bundles element by element.

2.10 BIT002: immediate retrieval interventions from identical states

BIT002 uses the frozen BIT/PL operator and 42,000 independent new factual HALF-law histories: 21,000 ALL_F and 21,000 ALL_M starts, a fixed equal mixture. Before generating a history, the checkpoint is selected uniformly from updates 193-256. Only factual CACHE history generates the stored pre-target state. At that state, recurrent target T_(t-2) and innovation target I_t are each evaluated with CACHE, ALL_FRESH and FOCAL_FRESH retrieval, giving six paired complete turnovers with the same semantic operator tape. Shadow outcomes do not feed into history. Every turnover retains the 32-parent, 128-query budget; target types and arms are paired observations, not independent replicates. This target comparison measures heterogeneity across specified targets, not the causal effect of changing environmental recurrence.

The pretreatment eligible set E contains policy-M parents with a valid cache; e is its size and H indicates that a focal eligible parent exists. ALL_FRESH replaces every eligible parent's cached policy probe with its paired fresh probe. FOCAL_FRESH changes only the eligible parent with the smallest prespecified 64-bit hashed key, with lower slot breaking ties, in an otherwise all-CACHE background; this is a hashed selection rule, not an exactly uniform focal-parent claim. The history, state, policy labels and common tape remain fixed by design. These are retrieval interventions conditional on the existing retention architecture, not interventions on the allele or cache retention.

In Sections 2.10 and 3.16-3.18, L/P/F and suffixes rec/inn are BIT002 symbols. In particular, BIT002 P_rec is an immediate recurrent-target contrast, whereas P_rec in Studies 001 and 003 is a long-run recurrence interaction; those estimands are not interchangeable. All six BIT002 contrasts are CACHE minus the corresponding FRESH intervention. L is eligible-parent descendant count divided by 32 under CACHE minus ALL_FRESH; P is normalized survivor accuracy under the same contrast, with denominator 1,024. F is the focal family's descendant count divided by 32 under CACHE minus FOCAL_FRESH. F is unconditional, including H=0 blocks as zero; neither e nor H is used to normalize effects. L and F involve different treatment vectors and cannot be merged or read as per-carrier effects.

Six frozen two-sided Hoeffding intervals use 42,000 blocks, range lengths 2 for L/P and 1/4 for F, alpha 0.012 for each L/P endpoint and 0.001 for each F endpoint (total 0.05), and exact upward-rounded rational half-widths 0.015608294208001326 and 0.002378115509733596. Joint coverage is at least 95% under the independent-uniform-Philox-word working idealization, which is not a mathematical or cryptographic proof of independence of generator outputs. Material thresholds are delta=1/64 for L/P and 1/256 for F; negligibility margins are epsilon=1/32 and 1/128. Exact fractions determine the ordered classifications before display rounding. This new

six-endpoint family is separate from every Study 001-003 family; there is no joint error control or pooled sample across studies. Supplementary Part D gives all estimates, prespecified transforms, histograms and audit limits.

3. Results

3.1 Study 001: execution, identity checks and record audit

The single production job (53530597) completed all 332,800 paths with exit code 0:0 and an elapsed time of 00:01:10. It made exactly 10,905,190,400 objective queries, 128 per update over 85,196,800 path-updates, with no N1, N2 or query-count failure. The frozen analyzer (job 53530688) and the separately implemented internal audit job (53530689) both completed with exit code 0:0. The accepted record lists 0 post-outcome changes.

Two audit tools, written separately from the producer and importing none of its code, checked the saved outputs. A C++ replay tool (MMEM-AUDIT-REPLAY-1.0.1) regenerated blocks 0-63 from block identifiers and keyed random streams and compared 64 blocks, 512 path records, 131,072 update records and 16,777,216 candidate-level audit rows against the saved outputs. After passing known-answer tests for SHA-256, Random123 Philox4x32-10 and multiply-high rejection, it reported PASS with zero mismatches. A Python verifier (MMEM-AUDIT-RECORDS-1.2.0) reconstructed all 22 saved estimates from saved path outputs, confirmed both SHAM identities in all 41,600 blocks, checked 332,800 path records and 85,196,800 update records, matched 97 manifest-listed files, recomputed the six frozen half-widths, and reported PASS with zero mismatches. These audits establish internal reproducibility of the recorded run. They were carried out within the project, partly with AI assistance; they are not external peer review and not an independent scientific replication.

3.2 Study 001: the start-state gate passed

Under HALF, the late-window ACTIVE frequency from ALL_M exceeded that from ALL_F by D_HALF = 0.000723, with familywise interval [-0.014898, 0.016343]. Under ZERO, D_ZERO = -0.000096, with interval [-0.015717, 0.015524]. Both intervals lie wholly inside (-1/32, +1/32); in percentage points, the two start contrasts were +0.072 and -0.010. Rule 2 therefore did not apply, and the enrichment rules were evaluated.

3.3 Study 001: recurrence strongly enriched the memory-use allele

The start-averaged late-window ACTIVE memory-use frequency was 0.810292 under HALF and 0.368657 under ZERO (Table 2, Figure 2). Relative to the exact neutral expectation, C_abs = 0.310292, with familywise interval [0.302482, 0.318102]: under recurrence the allele exceeded the neutral 1/2 baseline by 31.029 percentage points. Relative to the nonrecurrent law, C_rec = 0.441635, with interval [0.426015, 0.457256]: recurrence raised the late-window frequency by 44.164 percentage points. Both lower bounds exceed Delta = 0.03125 by a wide margin. Decision rule 3 applied, and the frozen primary decision is **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT**.

The individual cell means show the same pattern from both starts. Under HALF, ACTIVE frequencies were 0.809931 from ALL_F and 0.810654 from ALL_M. Under ZERO they were 0.368705 and 0.368609, below the neutral 1/2; this ZERO deficit is descriptive, because no inferential bound on A_ZERO - 1/2 was frozen. SHAM frequencies were 0.500083 and 0.499917 under ZERO and 0.499089 and 0.500911 under HALF; each pair sums to 1, as identity N2 requires.

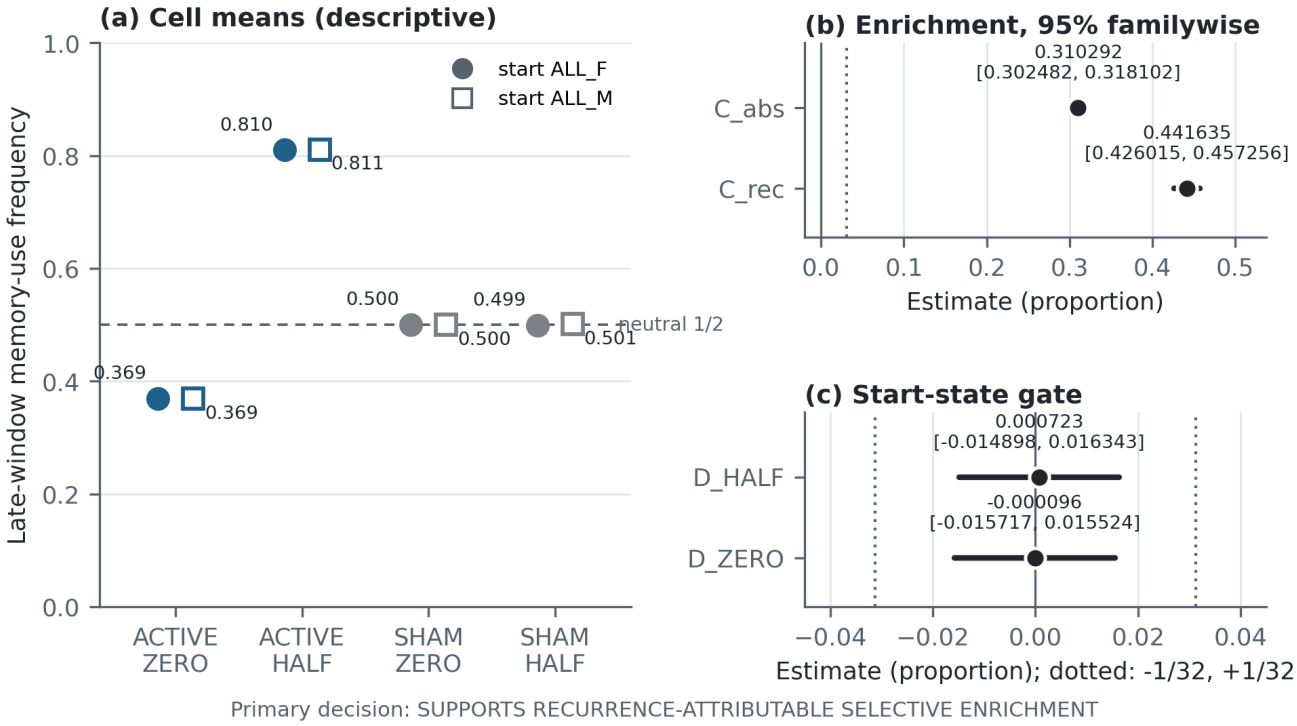


Figure 2. Study 001 allele-frequency outcomes. (a) Late-window memory-use frequency in all eight cells; markers distinguish the ALL_F and ALL_M starts, and the dashed line marks the exact neutral start-averaged expectation of 1/2. These cell means are descriptive. (b) The two enrichment estimands, C_abs and C_rec, and (c) the two start-state contrasts, D_HALF and D_ZERO, each with its frozen 95% familywise Hoeffding-Bonferroni interval; dotted lines mark the one-individual scale (+1/32 in b; -1/32 and +1/32 in c). Both enrichment estimands have lower bounds above +1/32, and both start-state contrasts lie inside the gate. All plotted values are read from the saved analyzer records at figure-generation time.

3.4 Study 001: population accuracy showed no resolved one-bit benefit

Under HALF, start-averaged ACTIVE accuracy was 0.628395, compared with 0.619018 for SHAM (Table 2, Figure 3). The recurrent-law accuracy contrast P_abs was 0.009377, with familywise interval [-0.005138, 0.023892]: an estimated contrast of +0.938 percentage points whose upper bound, 2.389 points, lies below the prespecified 3.125-point one-correct-bit threshold. The frozen classification is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**. Because this interval also includes zero, even the sign of the recurrent-law accuracy contrast is not resolved at the familywise level.

The recurrence interaction P_rec was 0.011360, with interval [-0.017669, 0.040390], or +1.136 percentage points with an interval from -1.767 to +4.039 points. The interval spans both zero and the one-bit scale, and the frozen classification is **UNRESOLVED**. The P_rec half-width, 0.02902925, is only slightly below Delta_P, so this design had limited capacity to resolve the interaction at the one-bit scale; the half-width was fixed in the frozen specification.

Under ZERO, start-averaged ACTIVE accuracy was 0.605327, compared with 0.607310 for SHAM: a small adverse descriptive contrast of -0.001983, or -0.198 percentage points. It carries no inferential bound and is retained as reported. SHAM accuracies were identical between starts (0.607310 under ZERO and 0.619018 under HALF), as identity N1 requires. The analyzer recorded the descriptive relation of P_abs to allele enrichment as DISAGREEMENT and that of P_rec as UNCERTAINTY.

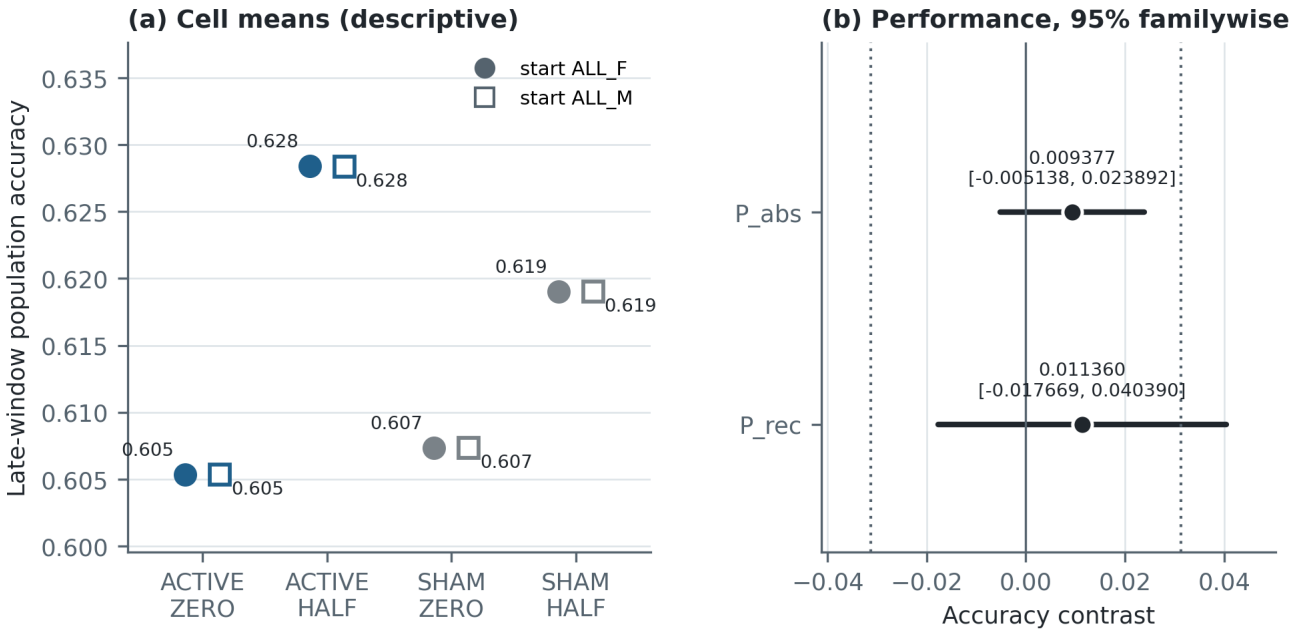


Figure 3. Study 001 population-accuracy outcomes. (a) Late-window mean population accuracy in all eight cells (descriptive); SHAM accuracies coincide between starts by identity N1. (b) The two secondary estimands with their frozen 95% familywise Hoeffding-Bonferroni intervals; dotted lines mark the one-correct-bit scale (-1/32 and +1/32). P_abs is bounded below the positive one-bit scale and P_rec is unresolved. All plotted values are read from the saved analyzer records at figure-generation time.

Table 1. Study 001 inferential estimands and frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary and secondary families.

Estimand	Estimate	95% familywise interval	Half-width	Frozen classification
C_abs	0.310292	[0.302482, 0.318102]	0.00781023	Lower bound exceeds 1/32
C_rec	0.441635	[0.426015, 0.457256]	0.01562046	Lower bound exceeds 1/32
D_HALF	0.000723	[-0.014898, 0.016343]	0.01562046	Inside (-1/32, +1/32)
D_ZERO	-0.000096	[-0.015717, 0.015524]	0.01562046	Inside (-1/32, +1/32)
P_abs	0.009377	[-0.005138, 0.023892]	0.01451463	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE
P_rec	0.011360	[-0.017669, 0.040390]	0.02902925	UNRESOLVED

Table 2. Study 001 descriptive late-window cell means (no inferential bounds were frozen for these 16 records).

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_F	0.368705	0.605324

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_M	0.368609	0.605330
ACTIVE, HALF, ALL_F	0.809931	0.628405
ACTIVE, HALF, ALL_M	0.810654	0.628385
SHAM, ZERO, ALL_F	0.500083	0.607310
SHAM, ZERO, ALL_M	0.499917	0.607310
SHAM, HALF, ALL_F	0.499089	0.619018
SHAM, HALF, ALL_M	0.500911	0.619018

3.5 Study 001: the joint result

Taken together, the primary and secondary families give one result rather than two. Lag-two recurrence strongly enriched a mutation-generated memory-use allele despite its fixed cost of one fresh proposal, both late-window start contrasts passed the prespecified finite-horizon ±1/32 start-state gate, and the enrichment was attributable to recurrence at the one-individual scale. Over the same paths, the recurrent-law ACTIVE-SHAM accuracy contrast was bounded below one correct bit per survivor with sign unresolved, and whether recurrence changed the accuracy effect of the allele was unresolved. In this model, selection on the use of a supplied cache was strong, while the ACTIVE-SHAM accuracy contrast was small at most and was not resolved at the prespecified scale.

3.6 Study 002: execution and record audit

The Study 002 production run (job 53553084) completed all 249,600 paths and made exactly 8,178,892,800 objective queries, 128 per update over 63,897,600 path-updates. The frozen analyzer (job 53553256) saved the 19 prespecified estimate records and the decision record, and a separately implemented audit (job 53553294) checked them. A C++ replay tool (PCONV-AUDIT-REPLAY-1.0.0) regenerated blocks 0-63 and compared 384 path records, 98,304 update records, 12,582,912 candidate-level audit rows and 16,384 permutation records with the saved outputs. Its known-answer tests included the Fisher-Yates and rejection-sampling construction, distance matching of the decoy at every displacement weight from 0 to 32, the arm-specific probe rule and the HALF law; it reported PASS with zero mismatches. A Python verifier (PCONV-AUDIT-RECORDS-1.0.0), which generated no random draws, reconstructed all 19 saved estimates, confirmed both SHAM identities in all 41,600 blocks, checked 249,600 path records and 63,897,600 update records with 128 queries at every path-update, matched 98 manifest-listed files and recomputed the seven frozen half-widths; it reported PASS with zero mismatches. As for Study 001, these are internal reproducibility audits, not external peer review or independent scientific replication.

An earlier Study 002 production submission failed before any computation because a required command-line argument, --design-go, was missing. It exited with status 64 and produced no production output, random draw, model execution or outcome. A prospective recovery addendum was recorded before the corrected run. This infrastructure-only event is documented in Supplementary Section S24.

3.7 Study 002: the start-state gate passed

D_INFO = 0.000129, with familywise interval [-0.015042, 0.015301], and D_NONINFO = -0.001268, with interval [-0.016439, 0.013903]; in percentage points the two start contrasts were +0.013 and -0.127. Both intervals lie wholly inside (-1/32, +1/32), so rule 2 did not apply.

3.8 Study 002: directional information had a bounded total effect on population accuracy

Start-averaged late-window accuracy was 0.628390 in INFO and 0.619740 in NONINFO (Table 4). The frozen primary estimate was Delta_P = 0.008650 (0.00864996836735652 at saved precision), with familywise interval [-0.006521, 0.023821] ([-0.006521316095375593, 0.023821252830088634] at saved precision). Directional cache information therefore changed late-window population accuracy by an estimated +0.865 percentage points, with an interval from -0.652 to +2.382 points. The upper bound lies below the 3.125-point one-correct-bit scale, the whole interval lies inside (-1/32, +1/32), and the interval contains zero. Decision rule 5 applied, and the frozen primary decision is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**; the sign of the effect is unresolved.

3.9 Study 002: directional information strongly amplified selection without a resolved absolute performance gain

Directional information strongly changed allele selection (Figure 4). Start-averaged late-window memory-use frequency was 0.810286 in INFO and 0.626083 in NONINFO, and exactly 1/2 in SHAM in every block. E_INFO = 0.184203, with familywise interval [0.169688, 0.198718]: directional retrieval left 18.420 percentage points (16.969 to 19.872) more memory-use carriers than the matched control. E_NONINFO = 0.126083, with interval [0.118825, 0.133340]: the matched control itself remained 12.608 points (11.883 to 13.334) above neutrality. Both are classified **MEANINGFUL POSITIVE AT THE ONE-BIT SCALE**.

Performance relative to the label-blind SHAM reference, whose start-averaged accuracy was 0.619009, did not follow. B_INFO = 0.009381, with interval [-0.005133, 0.023896], or +0.938 points (-0.513 to +2.390); B_NONINFO = 0.000731, with interval [-0.013783, 0.015246], or +0.073 points (-1.378 to +1.525). Both are classified **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE**, and neither sign is resolved. Because the lower bound of B_INFO does not exceed 1/32, Study 002 does not demonstrate a one-bit benefit of directional retrieval relative to no-memory use. The analyzer recorded the prespecified interpretive branch: "Directional information selects for policy use without converting into one-bit population performance, sharpening the selection-performance separation." These secondary classifications cannot alter the primary decision.

The cell means show the same pattern from both starts. INFO frequencies were 0.810221 from ALL_F and 0.810351 from ALL_M, and NONINFO frequencies 0.626717 and 0.625449. SHAM frequencies were 0.499678 and 0.500322, whose exact fractions sum to 1 as identity N2 requires, and SHAM accuracy was 0.619009 from both starts, as N1 requires.

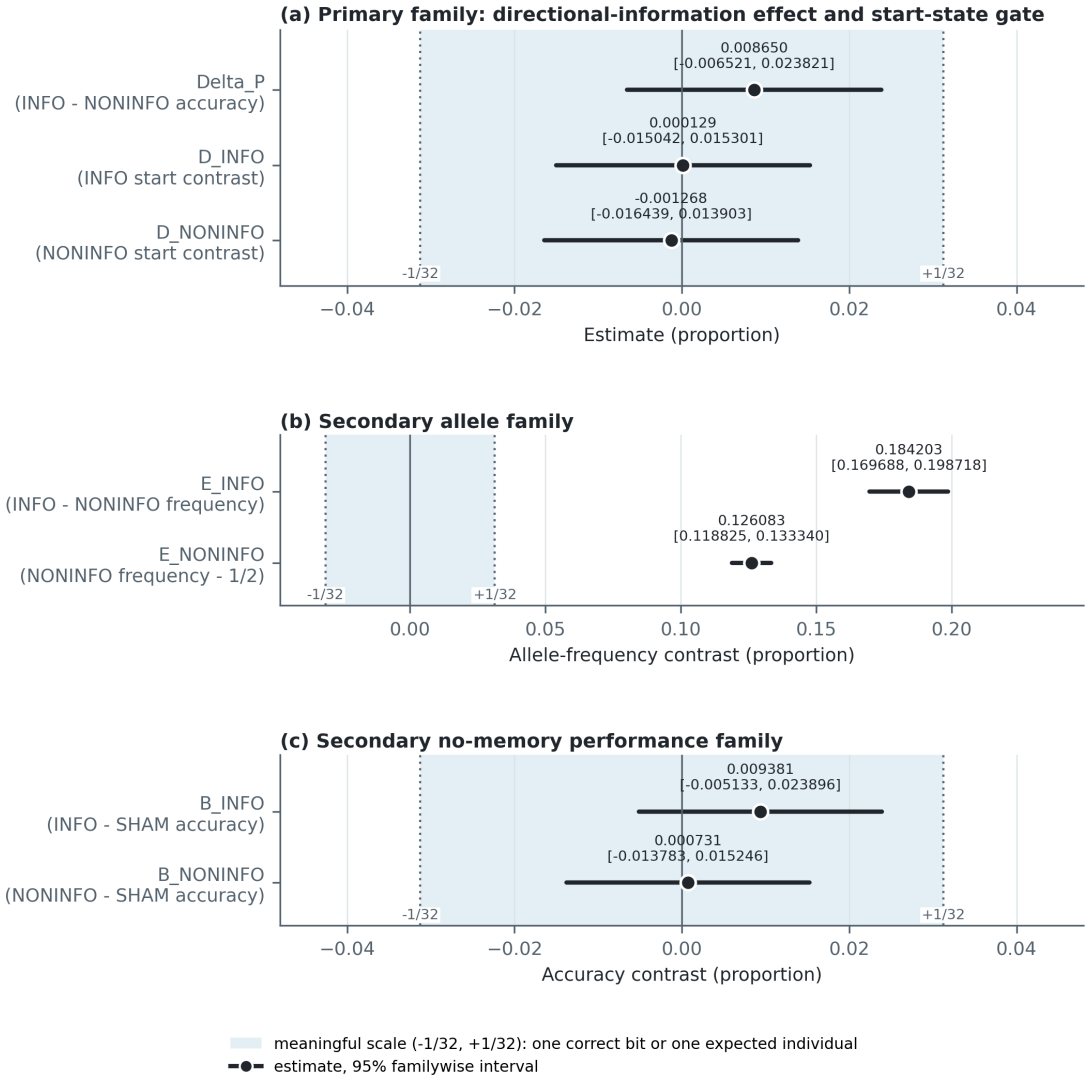


Figure 4. Study 002 directional-information outcomes. (a) The primary family: the directional-information effect on population accuracy, Delta_P (INFO minus NONINFO), and the two start-state contrasts D_INFO and D_NONINFO. (b) The secondary allele family: E_INFO (INFO minus NONINFO memory-use frequency) and E_NONINFO (NONINFO minus the exact neutral 1/2). (c) The secondary no-memory performance family: B_INFO and B_NONINFO, accuracy relative to SHAM. Each estimate is shown with its frozen 95% familywise Hoeffding-Bonferroni interval; the shaded band and dotted lines mark the meaningful scale (-1/32, +1/32), one correct bit in accuracy or one expected individual in allele frequency. Delta_P lies wholly inside the band, both allele contrasts lie far above it, and both performance contrasts have upper bounds inside it. All plotted values are read from the saved Study 002 analyzer records at figure-generation time.

Table 3. Study 002 inferential estimands and frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary, secondary allele and secondary performance families.

Estimand	Family	Estimate	95% familywise interval	Half-width	Frozen classification
Delta_P	Primary	0.008650	[-0.006521, 0.023821]	0.0151 7128	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE (rule 5; wholly inside ±1/32)
D_INFO	Primary (gate)	0.000129	[-0.015042, 0.015301]	0.0151 7128	Inside (-1/32, +1/32)

Estimand	Family	Estimate	95% familywise interval	Half-width	Frozen classification
D_NONINFO	Primary (gate)	-0.001268	[-0.016439, 0.013903]	0.01517128	Inside (-1/32, +1/32)
E_INFO	Secondary allele	0.184203	[0.169688, 0.198718]	0.01451463	MEANINGFUL POSITIVE AT THE ONE-BIT SCALE
E_NONINFO	Secondary allele	0.126083	[0.118825, 0.133340]	0.00725731	MEANINGFUL POSITIVE AT THE ONE-BIT SCALE
B_INFO	Secondary performance	0.009381	[-0.005133, 0.023896]	0.01451463	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE
B_NONINFO	Secondary performance	0.000731	[-0.013783, 0.015246]	0.01451463	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE

Table 4. Study 002 descriptive late-window cell means (no inferential bounds were frozen for these 12 records) and their start averages.

Cell	Memory-use frequency	Population accuracy
INFO, ALL_F	0.810221	0.628355
INFO, ALL_M	0.810351	0.628425
NONINFO, ALL_F	0.626717	0.619744
NONINFO, ALL_M	0.625449	0.619737
SHAM, ALL_F	0.499678	0.619009
SHAM, ALL_M	0.500322	0.619009
INFO, start average	0.810286	0.628390
NONINFO, start average	0.626083	0.619740
SHAM, start average	1/2 (exact in every block)	0.619009

3.10 The two-study result in the original operator bundle

The two prospective experiments in the original operator bundle give one result. In Study 001, lag-two recurrence strongly selected a costly memory-use policy while the recurrent-law accuracy contrast with SHAM was bounded below one correct bit with sign unresolved. In Study 002, under the same aligned law, directional cache information strongly amplified selection on the same policy: relative to a control that paid the same displacement cost and placed its candidate at the same distance from the parent, directional retrieval left 18.420 percentage points more memory-use carriers. Yet the total population-accuracy effect of that same information, mediated through evolved policy use, was bounded below the one-bit scale with sign unresolved, and neither INFO nor NONINFO showed a resolved accuracy gain over SHAM. Directional information therefore strongly amplified selection on the memory-use policy without a resolved conversion into population performance at the prespecified scale.

The 22 Study 001 records and the 19 Study 002 records are separate accepted estimate sets, 41 saved estimate records in total, from separate cohorts with separately controlled familywise families; they are not pooled. The Study 002 INFO arm applies the same retrieval rule as the Study 001 ACTIVE-HALF cells, and its descriptive start-averaged frequency and accuracy (0.810286 and 0.628390) are close to the Study 001 values (0.810292 and 0.628395). Study 002 extends the Study 001 producer and was neither designed nor analyzed as an independent replication, so this agreement is descriptive only.

3.11 Study 003: execution, audit-header defect and record audit

The single Study 003 production job (53561601) completed all 332,800 paths of the 41,600 blocks and made exactly 10,905,190,400 objective queries, 128 per update over 85,196,800 path-updates. Before any analysis or audit, a separate outcome-blind job (53562156) authenticated all 1,962 output files (4,694,355,040 bytes) against the production manifest and completion marker without parsing any scientific value. The frozen analyzer (job 53562267) saved the 22 prespecified estimate records and the decision record and recorded zero deterministic-identity failures.

The first audit attempt (job 53562358) stopped at an interface check. The authenticated production outputs had written chunk_index = 0 in the headers of the three non-chunk audit files (candidate rows, tournament-entry rows and per-update context rows), whereas the frozen binary schema required the non-chunk sentinel 0xFFFFFFFF; the producer's audit sink had constructed these headers with 0. By then the verifier had reconstructed all 22 estimates and checked every block, path and update record, but replay of the candidate, tournament and context payloads had not started. The attempt was preserved unchanged as a nonscientific interface failure and was not retried automatically. A prospectively frozen, normalization-only recovery then built a derived audit tree in which exactly bytes 48-51 of each of the three files were changed from 00000000 to ffffffff, 12 bytes in total, and proved that every payload byte from offset 64 to the end of each file was identical to the original; production outputs and analysis were not modified. The same accepted auditor then passed (job 53562452). Its C++ replay, which implements Threefry and circular loss separately from the producer and invokes no producer code, regenerated blocks 0-63 and checked 64 blocks, 512 paths, 131,072 path updates, 16,777,216 candidate rows, 16,777,216 tournament-entry rows, 131,072 context rows, 131,072 update rows, 512 path rows and 64 block rows. Its Python verifier checked all 41,600 block records, 332,800 path records and 85,196,800 update rows, confirmed N1 and N2 in all 41,600 blocks, reconstructed all 22 estimates and checked the audit payloads. Both reported PASS with zero mismatches. The original malformed files and the failed attempt are preserved. Release source corrects the producer so that non-chunk audit headers carry 0xFFFFFFFF; this one-line change touches no scientific operator, random stream, record or analysis, and no scientific rerun was warranted (Supplementary Section S37). As for Studies 001 and 002, these are internal reproducibility audits, not external peer review.

3.12 Study 003: the start-state gate passed

D_HALF = -0.000202, with familywise interval [-0.015822, 0.015419], and D_ZERO = -0.000240, with interval [-0.015861, 0.015380]; in percentage points the two start contrasts were -0.020 and -0.024. Both intervals lie wholly inside (-1/32, +1/32), so rule 2 did not apply.

3.13 Study 003: recurrence-attributable enrichment recurred in the alternative bundle

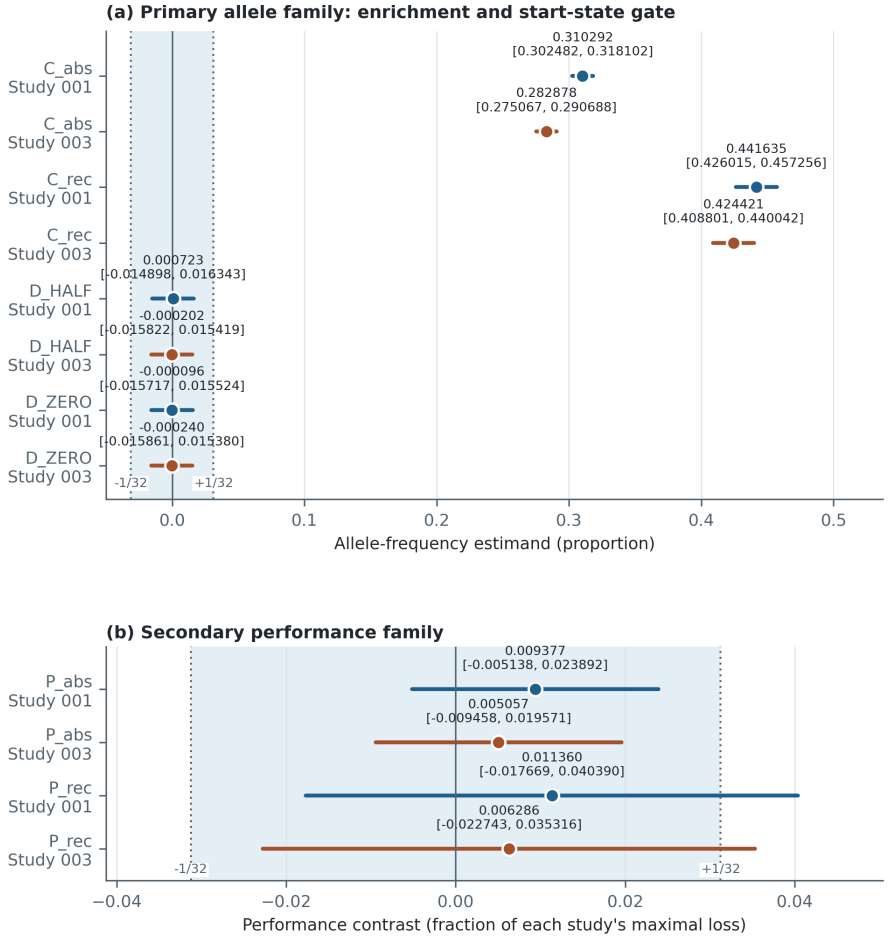
The start-averaged late-window ACTIVE memory-use frequency was 0.782878 under HALF and 0.358456 under ZERO (Table 6, Figure 5). C_abs = 0.282878, with familywise interval [0.275067, 0.290688], and C_rec = 0.424421, with interval [0.408801, 0.440042]: under recurrence the allele exceeded the neutral 1/2 by 28.288 percentage points, and recurrence raised its late-window frequency by 42.442 points relative to ZERO. Both lower bounds exceed Delta = 0.03125. Decision rule 3 applied, and the frozen primary decision is **SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT IN THIS ALTERNATIVE OPERATOR BUNDLE**. Table 5 gives every Study 003 inferential record at saved precision.

As in Study 001, the ZERO-law ACTIVE frequencies, 0.358576 from ALL_F and 0.358336 from ALL_M, were descriptively below 1/2; no inferential bound was frozen for this deficit. SHAM frequencies were 0.499608 and 0.500392 under ZERO and 0.499756 and 0.500244 under HALF, and each pair sums exactly to 1, as identity N2 requires.

3.14 Study 003: the performance contrast was bounded below the positive scale

Under HALF, start-averaged ACTIVE accuracy was 0.740842, compared with 0.735785 for SHAM. P_abs = 0.005057, with familywise interval [-0.009458, 0.019571], or +0.506 percentage points of the maximal loss (-0.946 to +1.957). Its upper bound lies below the declared 1/32 scale and its interval contains zero. The frozen classification is **BOUNDED BELOW THE POSITIVE ONE-BIT SCALE** (saved class BOUNDED_BELOW_POSITIVE; in Study 003 the label refers to the positive 1/32 performance scale of Section 2.9, not to a bit), with sign unresolved. P_rec = 0.006286, with interval [-0.022743, 0.035316], or +0.629 points (-2.274 to +3.532); it spans zero and the positive scale and is **UNRESOLVED**. These performance classifications cannot alter the primary decision. With primary support and P_abs bounded below the positive scale, the frozen crossed interpretation is **SELECTION/PERFORMANCE SEPARATION RECURS IN THIS OPERATOR BUNDLE**.

Under ZERO, start-averaged ACTIVE accuracy was 0.728441, compared with 0.729671 for SHAM: a small adverse descriptive contrast of -0.001230 (-0.123 points), in the same direction as in Study 001, with no inferential bound. SHAM accuracies were identical between starts, as identity N1 requires. Study 003 accuracy levels are not comparable with those of Studies 001 and 002: a uniformly random torus phenotype has expected accuracy close to 2/3 rather than 1/2.



Study 001: SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT; P_abs BOUNDED BELOW THE POSITIVE ONE-BIT SCALE; P_rec UNRESOLVED

Study 003: SUPPORTS RECURRENCE-ATTRIBUTABLE SELECTIVE ENRICHMENT IN THIS ALTERNATIVE OPERATOR BUNDLE; P_abs BOUNDED_BELOW_POSITIVE; P_rec UNRESOLVED

Figure 5. Study 001 and Study 003 at the declared scales. Each estimand is shown twice, once for Study 001 (32-bit genotypes, Hamming mismatch, exact weighted sampling without replacement) and once for Study 003 (32-locus 2^16-allele torus, graded circular loss, four-entry rank tournaments). The two studies use different operator bundles, separate implementations and independent cohorts; each has its own 95% familywise Hoeffding-Bonferroni families, and no estimate is pooled. (a) The primary allele family: C_abs, C_rec, D_HALF and D_ZERO; the allele scale 1/32 is one expected individual in both studies. (b) The secondary performance family: P_abs and P_rec, expressed as a fraction of each study's maximal loss; the 1/32 performance scale is one correct bit per survivor in Study 001 and 1/32 of maximal total torus loss in Study 003, so magnitudes are not comparable across studies and only the frozen within-study classifications are compared. The shaded band and dotted lines mark (-1/32, +1/32). In both studies both enrichment estimands lie far above +1/32, both start contrasts lie inside the band, P_abs has its upper bound inside the band and P_rec crosses +1/32. All plotted values are read from the saved analyzer records of each study at figure-generation time.

Table 5. Study 003 inferential estimands at saved precision and their frozen classifications. Intervals are two-sided 95% familywise Hoeffding-Bonferroni intervals over 41,600 blocks, computed separately for the primary allele family (alpha_each = 0.05/4) and the performance family (alpha_each = 0.05/2).

Estimand	Saved estimate	Saved 95% familywise interval	Half-width	Frozen classification
C_abs	0.2828775083101712740	[0.2750672787822218728, 0.2906877378381206753]	0.0078102 3	Lower bound exceeds 1/32
C_rec	0.4244214336688701923	[0.4088009746129713898, 0.4400418927247689949]	0.0156204 6	Lower bound exceeds 1/32
D_HALF	-0.0002015920785757212	[-0.0158220511344745237, 0.0154188669773230814]	0.0156204 6	Inside (-1/32, +1/32)
D_ZERO	-0.0002403963529146635	[-0.0158608554088134660, 0.0153800627029841391]	0.0156204 6	Inside (-1/32, +1/32)
P_abs	0.0050566998204140212	[-0.0094579258184457116, 0.0195713254592737541]	0.0145146 3	BOUNDED BELOW THE POSITIVE ONE-BIT SCALE; sign unresolved
P_rec	0.0062864162691967155	[-0.0227428350085227502, 0.0353156675469161812]	0.0290292 5	UNRESOLVED

Table 6. Study 003 descriptive late-window cell means (no inferential bounds were frozen for these 16 records). Accuracy is (2^40 - total survivor loss)/2^40 and is not on the Study 001 scale.

Cell	Memory-use frequency	Population accuracy
ACTIVE, ZERO, ALL_F	0.358576	0.728441
ACTIVE, ZERO, ALL_M	0.358336	0.728442
ACTIVE, HALF, ALL_F	0.782978	0.740842
ACTIVE, HALF, ALL_M	0.782777	0.740841
SHAM, ZERO, ALL_F	0.499608	0.729671
SHAM, ZERO, ALL_M	0.500392	0.729671
SHAM, HALF, ALL_F	0.499756	0.735785
SHAM, HALF, ALL_M	0.500244	0.735785

3.15 The three-study result across two operator bundles

The three prospective experiments give one paper-level result in two operator bundles. In the bit/Hamming bundle, recurrence strongly selected a costly memory-use policy (Study 001), and directional cache information strongly amplified that selection (Study 002), without a resolved one-bit gain in population accuracy. In one independently implemented torus/tournament bundle with new cohorts (Study 003), the frozen primary classification of Study 001 recurred, the recurrent-law ACTIVE-SHAM performance contrast was again bounded below the positive 1/32 scale with sign unresolved, and the recurrence interaction in performance was again unresolved; the frozen crossed interpretation is that the selection/performance separation recurs in this operator bundle. This is agreement between prespecified within-bundle classifications. Because alphabet, loss, proposal and resampling semantics, and survival all differ between the bundles, Study 003 shows that the separation is not confined to the original bundle, but it does not show that the separation holds in every bundle, and it cannot say which change, if any, matters.

The 22 Study 001, 19 Study 002 and 22 Study 003 records are three separately accepted estimate sets, 63 saved estimate records in total, each with its own familywise families; none is pooled with another. Effect magnitudes are not compared across bundles, because the performance scale has a different meaning in Study 003 and its accuracy levels sit on a different baseline.

3.16 BIT002: execution and independent internal audit

FRCE Slurm job 54269109 completed the full 42,000-block experiment on October 25, 2026, in 34 seconds; all original job and program steps succeeded. The original analyzer and separately written auditor produced byte-identical results. Independent execution-audit acceptance preceded the first unseal of estimates. An earlier scientific job, 54254461, failed in its runtime output monitor, produced no accepted estimates and remains INVALID; its partial files were neither resumed nor combined with R2. One fresh R2 attempt followed explicit authority, a bounded runtime correction, independent source review, synthetic FRCE engineering validation and renewed readiness gates. Supplementary Section S44 preserves the failed execution and the accepted audit's limits. Internal implementation, execution and publication-content reviews are not journal peer review or independent scientific replication.

3.17 BIT002: complete immediate-effect results

Table 7. BIT002 immediate effects in percentage points of each normalized endpoint, all 42,000 blocks. Joint coverage is at least 95% for these six intervals under the stated independent-uniform-word idealization. L/P compare CACHE with ALL_FRESH; F compares CACHE with FOCAL_FRESH. These are not late-window allele-frequency or ACTIVE-minus-SHAM estimands.

Endpoint	Estimate (pp)	Simultaneous interval (pp)	Frozen classification
L_rec	5.041741	[3.480912, 6.602570]	MATERIAL_POSITIVE
L_inn	-0.752009	[-2.312838, 0.808820]	NEGLIGIBLE_WITHIN_MARGIN
P_rec	2.484087	[0.923257, 4.044916]	POSITIVE_MAGNITUDE_UNRESOLVED
P_inn	-0.571308	[-2.132137, 0.989522]	NEGLIGIBLE_WITHIN_MARGIN
F_rec	1.239360	[1.001549, 1.477172]	MATERIAL_POSITIVE
F_inn	-0.194271	[-0.432082, 0.043541]	NEGLIGIBLE_WITHIN_MARGIN

At recurrent targets, L and F lower bounds exceed their respective material thresholds, 1.5625 pp and 0.390625 pp. P_rec is positive, but its interval crosses the material threshold 1.5625 pp and the negligibility margin 3.125 pp; its magnitude remains unresolved. All innovation point estimates are negative and their intervals contain zero, while lying within their respective negligibility margins (3.125 pp for L/P; 0.78125 pp for F). These bounds do not establish null effects or absence of adverse effects within the intervals. The exact algorithmic summary is S3_MATERIAL_POSITIVE_ONLY endpoints=L_rec,F_rec; it is not a favorable-mechanism label.

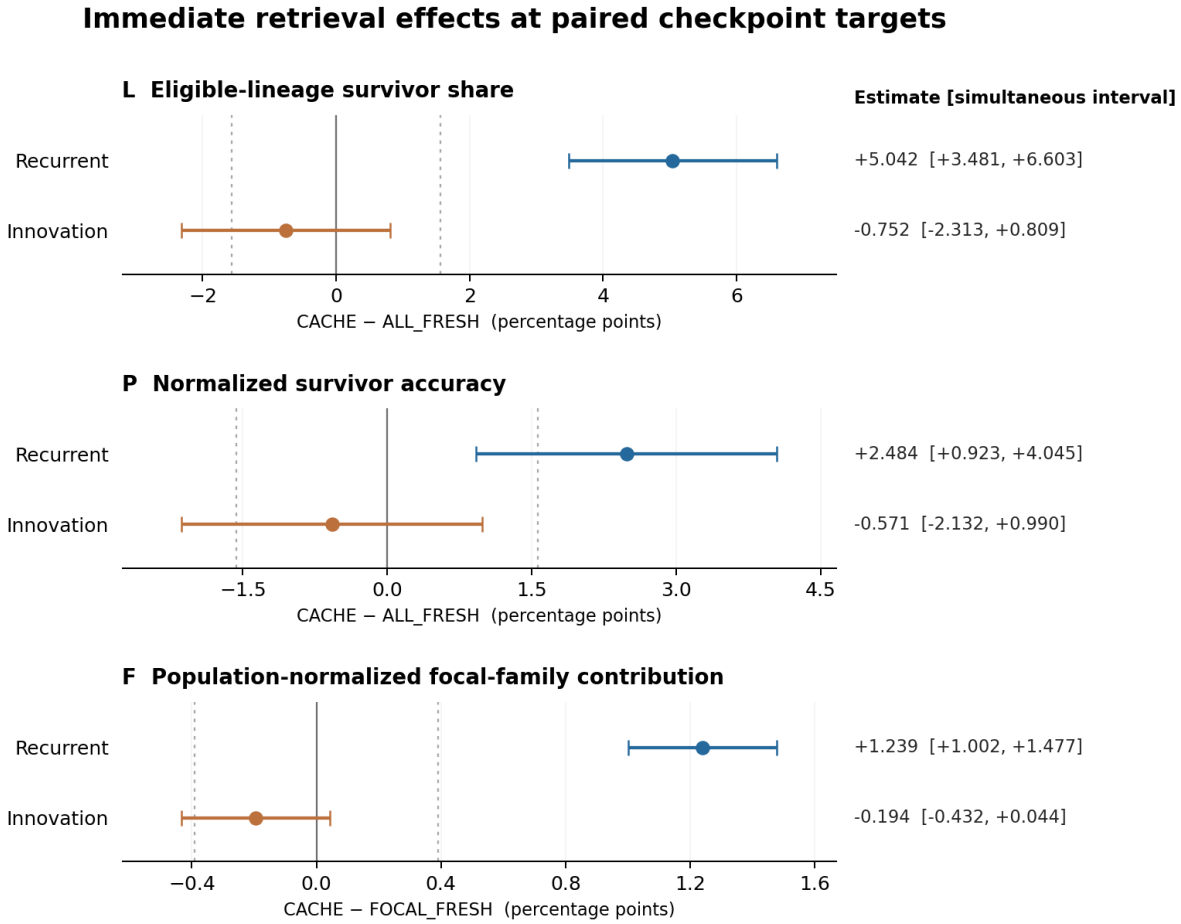


Figure 6. All six BIT002 immediate retrieval effects, copied without alteration from the accepted publication update. Panels distinguish lineage share, normalized survivor accuracy and unconditional focal-family contribution. L/P use CACHE-minus-ALL_FRESH; F uses CACHE-minus-FOCAL_FRESH. Points and intervals use percentage points; dotted lines mark the panel's positive and negative material thresholds. Intervals jointly cover the six means with probability at least 95% under the independent-uniform-Philox-word working idealization. Exact classification precedes display rounding. F is population-normalized, not per carrier. Bounded innovation effects are not established nulls. Full exact results and the original caption are retained in the accompanying evidence.

3.18 BIT002: prespecified transforms and structural qualifications

The recurrent-minus-innovation interval is [2.672091, 8.915409] pp for L and [0.958008, 1.909254] pp for F, both excluding zero, but [-0.066264, 6.177053] pp for P, which includes zero. A positive recurrent P interval and a zero-containing innovation interval therefore do not establish a resolved P difference between targets. These prespecified transforms describe effect heterogeneity, not an intervention on recurrence. The average-target intervals are [0.584037, 3.705695] pp for L, [-0.604440, 2.517219] pp for P and [0.284733, 0.760356] pp for F; they are fixed interval transforms, not additional tests.

H=1 occurred in 20,996/21,000 ALL_F and 20,995/21,000 ALL_M blocks; nine blocks had H=0. Exactly 2,551/42,000 blocks had e=32. The structural bound is abs(L) <= min(1, e/8, (32-e)/8), with L forced to zero at e=0 or e=32, and F zero at H=0. Bounded aggregate lineage effects must be read alongside that constraint and the complete 33-bin eligibility histograms in Supplementary Section S43. No e/H-normalized, stratified, conditional-ratio or additional inferential result was introduced.

4. Discussion

The central observation is a separation between what selection favored and what the population gained. In Study 001, under the aligned recurrent law, more than four-fifths of late-window survivors carried the memory-use allele in both start conditions over the 256-update horizon, whereas the ACTIVE-SHAM accuracy contrast, in which SHAM labels are inert, was bounded below one additional correct bit per survivor. Under the nonrecurrent law, where the cache carried no lag-two information, the allele fell descriptively below its neutral frequency, and ACTIVE accuracy was descriptively slightly below SHAM.

Study 001 was not designed to identify the mechanism of this separation, and the following explanations are hypotheses for future work rather than findings. Survival depends on the relative scores of candidates within an update. A carrier whose cache matches a recurrent target supplies a candidate that outscores most fresh proposals, so carriers can be strongly overrepresented among survivors even if the survivor set, which is dominated by the best candidates from all families, improves only modestly on average. At non-recurrent updates the same carriers have given up one fresh draw, and the local child of every parent already supplies a near-copy of the best current candidate, which may limit what a retrieved record adds to the population's mean fit. The ZERO accuracy contrast is in the direction expected from displaced exploration, but it is small and descriptive.

Study 002 sharpens this separation rather than reversing it. Its control removes the direction of the retrieved displacement at recurrent updates while keeping the same displacement cost and the same distance from a parent that has just survived selection. Relative to that control, directional information added a large and well-resolved selective advantage to the memory-use policy, but the total population-accuracy effect of the same information was bounded between -0.652 and +2.382 percentage points. Two further observations are descriptive and motivate hypotheses only. First, NONINFO was itself selected well above neutrality; one possibility is that a candidate placed at the cache's distance from a recently selected parent tends to score better than a uniform fresh proposal even without directional information, particularly because a large share of displacements had weight 0 and so reproduced the parent (Supplementary Section S23). Second, neither arm showed a resolved accuracy gain over SHAM, consistent with the Study 001 mechanisms suggested above, in which strong relative advantage among candidates can coexist with a small change in the survivors' mean fit. Study 002 was not designed to test these explanations.

Study 003 asks whether the separation belongs only to the original operator bundle. Its bundle replaces the binary alphabet with 2^16 circular alleles per locus, Hamming mismatch with graded squared circular loss, XOR-mask proposals and bit flips with uniform vectors and coordinate resampling, and exact weighted sampling without replacement with rank-only tournaments in which even the unique best candidate leaves no descendant with probability above one-third; it was implemented anew with new cohorts. Recurrence-attributable enrichment of the same supplied-cache-use policy recurred, and the recurrent-law performance contrast was again bounded below the declared positive scale with sign unresolved. The selection/performance separation is therefore not specific to Hamming loss or to near-deterministic preservation of the best candidate, but Study 003 was not designed to test the mechanisms suggested above, and because all of its operators changed together it cannot attribute either the recurrence of the classification or any difference in magnitude to a particular component.

BIT002 establishes immediate retrieval effects at identical stored states in one bundle: cached retrieval materially increased eligible-lineage share and the focal family's population contribution at recurrent targets, and increased normalized survivor accuracy with unresolved magnitude. This does not contradict the bounded or unresolved long-run performance results in Studies 001-003. The interventions differ (CACHE versus paired FRESH at one state, versus diverging ACTIVE/SHAM or INFO/NONINFO trajectories), as do the time horizon, state distribution and endpoints. A short-term accuracy benefit can coexist with bounded long-run arm differences; no quantitative mediation or offsetting-exploration account follows from that coexistence. In particular, L/F are descendant-share effects, not a demonstrated long-run allele effect, and F does not estimate an individual's average private benefit. The proposed candidate-competition explanations above remain hypotheses.

Only factual-history blocks 0-63 were independently replayed in BIT002. For the other 41,936 blocks, provenance rests on fixtures, that replay and check E rather than full history replay. A forged but self-consistent pre-target state with consistently recomputed arms can pass checks C-J by design: conditional replay verifies transitions from a stored state without proving its historical validity. These scope limits qualify the identical-state interpretation. The protocol's prospective directional intuition receives no hypothesis, test or interpretive privilege, and is not used to label the findings expected or mechanistic confirmation.

These results are consistent in spirit with prior findings that the marginal benefit of learning or memory can be small even where memory is favored [4], and with models in which memory-related loci or anticipatory effects evolve only under particular environmental regimes [5, 7]. The present design adds an exact neutral benchmark for a mutable use allele, a cost expressed in the same currency as search itself, and prospectively separate scales for enrichment and performance. Its main methodological implication is that an allele-frequency response to environmental structure, however strong, should not be read as a population-performance response without a separately stated performance estimand and scale. Study 002 extends that implication to the information content of the retrieved record itself: even an allele-frequency effect attributable to directional information, measured against a control matched on displacement cost and parent distance, need not convert into a commensurate population-performance effect. Study 003 shows that the implication is not tied to one operator bundle.

The result is deliberately narrow. HALF matches the recurrence lag to the retrieval delay and so favors retrieval by construction; no other lag or copy probability was tested in either bundle. The opportunity cost is specific to replacing one effectively uniform proposal in a four-candidate budget. The cache architecture was supplied rather than evolved. The Study 002 control matches

parent distance and one common coordinate relabeling, not every relation among genotypes, and its primary estimand is a total effect through evolved policy use rather than a per-use value of information. Study 003 adds one alternative operator bundle, not a family of bundles, and its proposals are uniform rather than gradual. Within those conditions, the long-run allele inferences are strong and the long-run performance inferences are bounded or unresolved. BIT002 adds sign-resolved immediate recurrent-target accuracy and material lineage/focal effects, but does not extend those conclusions beyond its one-turnover BIT/PL setting.

5. Limitations

- **One aligned recurrent law.** HALF was a prior, outcome-informed choice that matches the cache delay by construction, and Study 002 used HALF alone. Study 003 used the same ZERO and HALF laws in its own bundle. No other lag, copy probability or target law was tested, and the result does not establish selection under arbitrary recurrence.
- **One fixed cost per bundle.** The cost is the replacement of one of two effectively uniform fresh proposals. It is a model-specific search-budget trade-off, not a calibrated energetic or biological cost and not a general cost scale. No other cost was tested in either bundle.
- **One mutation rate and one horizon; two operator bundles.** All three studies use mutation rate 1/32 and 256 updates. Studies 001 and 002 share one abstract model class; Study 002 extends the Study 001 producer and is not an independent model-class reproduction. Study 003 adds one independently implemented alternative operator bundle, which is neither a sample from nor a test of operator bundles in general.
- **One alternative bundle; no component attribution.** Study 003 changed the alphabet, the loss, the proposal and resampling semantics and the survival rule together. Its result cannot be attributed to any one of them, and it does not establish universal replication, continuous or gradual search (its proposals are uniform and its local operator resamples coordinates rather than taking small steps), or the outcome under any other tournament size.
- **Supplied architecture.** The one-record private cache and its retrieval rule were supplied. Only the use policy mutated. No study addresses the spontaneous origin of memory architecture or capacity.
- **Retention and retrieval are not separated.** In all three studies the allele jointly determines whether the one-step record is retained and whether it is retrieved; every individual has a cache slot, but only M carriers receive a valid record, so this design cannot separate selection on retention from selection on retrieval.
- **A total, not per-use, information effect.** Delta_P is the total population-accuracy effect of directional cache information beyond the parent and matched displacement length, mediated through evolved policy use. INFO and NONINFO diverge in policy frequency, cache use and survivors, so Delta_P is not a per-use value-of-information estimate and supports no per-use causal claim.
- **Control geometry.** The shared permutation preserves each decoy's distance to its parent and one common coordinate relabeling of displacement masks, but not every cross-genotype relation. Displacements of weight 0 or 32 cannot be scrambled and retain direction. In NONINFO, 42,175,734 of 100,158,252 decoy applications from ALL_F and 45,696,359 of 108,736,742 from ALL_M had weight 0 (42.1% and 42.0%); none had weight 32.
- **Finite horizon; start-state gate only.** In all three studies, both late-window start contrasts were bounded within one expected individual (95% familywise) over 256 updates. This is not a demonstration of general start-independence, stationarity, equilibrium, evolutionary stability, rare-mutant invasion or fixation probability.

- **Performance not resolved at the one-bit scale.** P_abs excludes a gain of one correct bit but does not resolve its sign; P_rec is unresolved, and its half-width is close to the one-bit scale. In Study 002, Delta_P, B_INFO and B_NONINFO exclude a positive one-bit effect but none resolves its sign. In Study 003, P_abs excludes a positive effect at its own 1/32 scale without resolving its sign, and P_rec is unresolved. A long-run population benefit at the declared scale was not demonstrated in Studies 001-003. BIT002 instead resolves a positive immediate accuracy effect at recurrent targets, with magnitude relative to its material and negligibility thresholds unresolved; it does not establish long-run population benefit. The small adverse ZERO accuracy contrasts and the ZERO allele deficits in Studies 001 and 003 are descriptive.
- **BIT002 scope and provenance.** BIT002 is restricted to late HALF-generated BIT/PL states, an equal start mixture and one turnover. It supplies no TORUS retrieval-intervention result, long-run mediation, intervention on recurrence, e/H-normalized effect or per-carrier claim. Its six intervals rely on the independent-uniform-word idealization; only 64 histories were independently replayed, and a self-consistent forged state can pass conditional checks. Near-monomorphism structurally constrains L. These limits remain after the internal audit passed.
- **Separate families and separate estimate sets.** Each of the two Study 001 families, the three Study 002 families and the two Study 003 families has 95% familywise coverage; coverage is not controlled jointly across families, across studies or across the 44 descriptive means. The 22 Study 001, 19 Study 002 and 22 Study 003 records are three separately accepted estimate sets and are not pooled or presented as independent replications. The 42,000 BIT002 blocks and six-endpoint family are additional separate evidence; no combined sample size or manuscript-wide 95% coverage is claimed.
- **What SHAM does not control.** SHAM is an exact neutral benchmark for label dynamics, not a control for alternative caches, alternative costs or alternative retrieval rules. ACTIVE and SHAM genotypes and genealogies diverge at the first valid-M retrieval, so ACTIVE-SHAM contrasts are arm-level comparisons, not path-level counterfactuals; the same holds for INFO, NONINFO and SHAM in Study 002 and for ACTIVE and SHAM in Study 003.
- **Abstract model.** The genotype, objective, selection rule and population are abstract. No biological, ecological, neural or engineering extrapolation and no claim of general population utility is supported, and the study does not replicate earlier internal private-memory studies.
- **Internal audit only.** Replay and record verification were performed by separately written tools within the project, partly with AI assistance. They are not external peer review or an independent scientific replication. Producer and audit tools were both written with AI assistance from the same specification; the audits detect implementation divergence, not a shared misreading of the specification. In Study 003 the first audit attempt stopped at three malformed audit-file headers; the recovery changed 12 header bytes in a derived audit tree only, and the original files and the failed attempt are preserved.
- **Frozen studies and separate prospective extensions.** No accepted study is extended, topped up, retuned or relabeled in this revision. BIT002 was a separately frozen prospective experiment and does not complete the broader PHASE3-SELECTION-PERFORMANCE-MECHANISM-001 protocol. Remaining scientific questions require separate prospective designs; this manuscript integration authorizes no new computation.
- **Bounded literature search.** The literature comparison rests on a structured search of open scholarly metadata with targeted citation chaining, not an exhaustive systematic review; it supports a bounded distinct-contribution claim only and cannot establish a unique nearest precedent. The search archive will be included in the public package (Section 7).

6. Methods summary

For Study 001, the model, cohorts and analysis follow the frozen specification PHASE2-MUTABLE-MEMORY-001 revision 1 exactly; Supplementary Sections S2-S10 give the full update order, target laws, allele mutation, cache inheritance and use, candidate budget, selection, estimands, interval construction and decision rules. Random values were drawn from canonical Random123 Philox4x32-10 streams whose keys are derived from SHA-256 of a purpose-specific namespace string and whose counters are the coordinates (block, update, entity, subindex); a coordinate-schema enumeration and collision audit preceded production. Ten deterministic fixtures, including hand traces, memory-better and memory-worse cases, duplicate and invalid-cache cases, full SHAM identity paths, target-law pairing, an exact weighted-selection fixture with a forced rejection, random-coordinate invariance and single-path regeneration, passed before production; none contains a sampled scientific outcome.

Per-update binary records were saved for every path, per-path late-window summaries and final-state hashes for every path, and full candidate-level rows for the fixed audit blocks 0-63. The frozen analyzer computed the 22 estimate records, their bounds and classifications from saved path outputs. Interval endpoints use exact rational estimates and high-precision half-widths. Descriptive diagnostics (mutation counts, cache-probe use and survival, and occupancy counters) were saved for interpretation only (Supplementary Section S12).

Study 002 follows the frozen specification PHASE2-PERFORMANCE-CONVERSION-002 revision 1 exactly; Supplementary Sections S16-S26 give the arm definitions, shared-permutation construction, event order, estimands, interval families, frozen decision order, all 19 records, diagnostics and audit. Random values come from the same Philox construction under the new namespace PHASE2-PERFORMANCE-CONVERSION-002|production-r1|<purpose>, with one added purpose, DECOY_PERMUTATION, generated for every block and update regardless of branch or arm. The specification required ten deterministic fixtures before production, covering bit-identity of the unchanged operators, Fisher-Yates and rejection fixtures with forced retries, INFO-NONINFO identity through innovation updates, distance matching at all 33 displacement weights, preservation of weights and overlaps under the shared permutation, the uniform subset law of the decoy, full SHAM identity paths, random-coordinate invariance and single-path regeneration; none contains a sampled production outcome. Per-update records, per-path summaries and final hashes, the 19 estimate records and full candidate rows and permutations for audit blocks 0-63 were saved. The frozen analyzer computed all estimates from saved path outputs in exact rational arithmetic with high-precision half-widths.

Study 003 follows the frozen specification PHASE2-TORUS-MEMORY-003 revision 1 exactly; Supplementary Sections S27-S39 give the exact torus loss, proposal construction, tournament and tie semantics, event order, Threefry key, counter and lane addressing, SHAM identities, estimands, interval families, decision order, all 22 records, execution and audit provenance and exact claim limits. Random values come from canonical Random123 Threefry4x64-20, pinned to one official reference commit whose known-answer vectors both producer and auditor were required to pass. Each purpose's four 64-bit key words are taken from SHA-256 of PHASE2-TORUS-MEMORY-003|production-r1|<PURPOSE>, every counter is (block, update, entity, subindex), and each 32-coordinate vector is read from two calls as sixteen 16-bit lanes per call. The specification required sixteen deterministic fixtures before production, including the Threefry known-answer vectors, a complete address enumeration and collision audit, circular-distance wrap and antipode cases, maximal-loss cases, one- and two-update hand traces, donor, tournament and candidate-key tie cases, the exact 128-rank tournament table, full SHAM paths and single-path regeneration; none samples a scientific block. Per-update records, per-path

summaries and final hashes, per-block variables with N1/N2 flags, full candidate, tournament-entry and context rows for audit blocks 0-63 and the 22 estimate records were saved. The frozen analyzer computed every estimate in exact rational arithmetic from saved block numerators, with high-precision half-widths.

BIT002 follows accepted protocol v0.4, SHA-256 71ccc1d065fa4868c30f96588ec34312bdc435e1047214c60b1f5643846c7daa. Producer and auditor sources, own auditor FIXOUT and binaries were frozen before cross-implementation output access; corruption and agreement fixtures preceded the outcome-blind resource qualification and single benchmark. The original R2 analyzer and auditor computed all six effects once from the accepted run. This revision copies accepted results and figures; it performs no simulation, producer/analyzer/auditor rerun, new raw-record analysis or statistical test. Supplementary Part D records the exact lineage, accuracy and focal contrasts and the narrower provenance guarantees.

This is an abstract computational model. It involves no human participants, no animals and no personal data.

7. Data and code availability

The accepted v1.2 package and its recorded release provenance are preserved. Version 1.3.0 adds BIT002 manuscript text, an unchanged accepted figure and a reproducibility release on the existing GitHub repository. This is a research preprint, not an externally peer-reviewed publication. The archived Studies 001-003 package contains the frozen specification, source code, configuration, build environment and SHA-256 manifests; the saved per-update, per-path and per-block outputs; candidate-level audit rows; analyzer records; internal audit receipts; the prospective design-review, production-authorization, production-submission and acceptance records; the literature-search archive (search script, API responses, deduplicated records and title screen); and this manuscript source with its figure and statistics-checking scripts. Manuscript text, figures and data will be released under the Creative Commons Attribution 4.0 International license (CC BY 4.0), and original code under the MIT License. The versioned manuscript, source and new BIT002 reproducibility assets are provided at https://github.com/jackchenx3/recurrence-selects-costly-memory-use/releases/tag/v1.3.0. Earlier Studies 001-003 archives remain available under their prior versioned releases and Zenodo DOI 10.5281/zenodo.23085659; that DOI identifies version 1.2.0, not this revision. The authenticated outputs used here have the following SHA-256 values: decision record 5b3aea3c18c4530b2be9f29d5c64fdd8173b6b323823bf9cc3361bd4e7010fc0; 22 estimate records 83828a49936bb8364813e2634f11cb28c4de2212b4b162d96e616ddbc7b4f8c0; descriptive diagnostics 51554620a5e302416edf8de8a073c6962abbc2c8967da80b91b83e4c3d5b685c; replay receipt 9b40322d0d09b90f0a6862533c1efdc7aff21bd699a882b9b9555c5f7c4b4857; record-verification receipt afd96dc0a5b190401c035fc66c8af8fd4d4603e7185d05bff73fb05e698cff5a; production inventory audit 622fa9e01a40c44ba545926e54b06e102d20b7fcf9e9c8e645ddaf9f7212e09c.

Prospective timing is recorded as follows (all times UTC, 2026-10-01). The frozen specification, design/PHASE2-MUTABLE-MEMORY-001-REV1.md (revision 1), has SHA-256 c17a3a1e9ac9cf2260f6743eaf08e4c2808080a767af16d2f9e9d2b9ad3e821d. Its terminal design review by a non-implementing AI reviewer (record CLAUDE-003, verdict GO) was completed at 01:55:25; the review has SHA-256 787cf32a20312f2682c761243f3dec04c5eb8036d14d6efc444e9830db37960a and its receipt 6277757f9ab3dfa611515c161776c17b561738eba6fa478ae0f8634e13c03176. The production execution authority (SHA-256

ee3cb68bed5d212e6ea10ece542e2e90c32cb9cea07d85fceda10fc56b375b1d) was issued at 12:53:04, and the sole production job (53530597) was submitted at 12:55:19 with no automatic retry and no parameter changes. The acceptance record has SHA-256 2eb3aaf5d754f4570a75522aa9e1f38358715c06c674d5a267732a1357926e6f. The frozen design and its GO review therefore precede the only production authorization and submission.

Study 002 is a post-v1.0 prospective extension added in version 1.1.0 of this manuscript; the Study 001 values, records and provenance above are unchanged. The archived package also contains the Study 002 frozen specification, producer extension, configuration and manifests, saved per-update, per-path and per-block outputs, candidate-level audit rows and permutations for blocks 0-63, analyzer records, internal audit receipts, the record of the failed original production submission, the prospective recovery addendum and the acceptance record, under the same licenses. The authenticated Study 002 outputs have the following SHA-256 values: decision record 3f28b82d0cb40357a1068c464d1c5158c36ba8fbfe775026ce22c70113b3b84a; 19 estimate records 5efa213cdfeb985a20fd7bbc65fb36dd505c568670ad350b3223c2139eb0b68d; descriptive diagnostics 304843eb5d4a38761a6ce371c7cc125260041dfd910b29d5ba3cc377a6ac3553; replay receipt cbee79527876745e8918576f73a5d302377e4463d959318ca2f5feffa66416d4; record-verification receipt 09f5dccf4921c8d842ef60fb31aca712bbb582e87a2011c76c02efb658539197; production run manifest b7aceaf6638e8f06246c8fb9b182c280a8ed98c4444d93a4a60decdbafb80136; acceptance record fc7777239cd4442c35c01d6261fd71e6fbf7e7f6fd6e11ddcc5d24876ce2f4a2 (accepted at 07:10:51 UTC on 2026-10-02). The frozen Study 002 specification, design/PHASE2-PERFORMANCE-CONVERSION-002-REV1.md (revision 1), has SHA-256 58ee414d753fab042e6cccee1a4e6b95dd7f167633b4f33c1491687d9afcb895. Its non-implementing terminal review returned DESIGN_GO; the review has SHA-256 76b0c6172e61f60befecef183e83d9e43ea50c1b6bc1e93299668b76f55be60f, and DESIGN_002_GO.json (SHA-256 f4a6180109e0d212cbe784a7722a64576e5e7cb0fbfd4b23c757bb5e7ab84d6a) was frozen at 18:59:00 UTC on 2026-10-01. The original production authority (SHA-256 31c2100dd73de0090ded3059525aacd139565917086d2c19880b01e319c98992) was issued at 06:02:41 UTC on 2026-10-02. After the infrastructure-only failure, the failure record (SHA-256 289bf3188286476698136d499c12eb3a8c9e4e6fc5cde18fcb15718395f4afaa) was accepted at 06:28:45, the prospective recovery addendum (SHA-256 db12c0143b6ba2c0b8cd3771ad18b9afe12feb069fe2a97d15227f4fad836e83) was issued at 06:29:59, and the one-recovery execution authority (SHA-256 c4cb2e4c162df92a761b2fff391b6e81e000055db88d7cbaeb954b0750b4e085) was issued at 06:32:55, all UTC on 2026-10-02.

Study 003 is a post-v1.1 prospective extension added in version 1.2.0 of this manuscript; the Study 001 and Study 002 values, records and provenance above are unchanged. The archived package also contains, under the same licenses, the Study 003 frozen specification and design arithmetic; the pinned Random123 reference provenance; the separately written producer, analyzer and auditor sources, configuration and manifests; the saved per-update, per-path and per-block outputs; the candidate, tournament-entry and context rows for audit blocks 0-63, including the three original audit files with their malformed headers; the outcome-blind production-output authentication; the preserved failed audit attempt; the normalization receipt and derived-audit manifest; the recovery audit receipts; the acceptance record; and the post-audit source-patch receipt. The authenticated Study 003 outputs have the following SHA-256 values: decision record 84f88bb17c20593abd222b8e099e2981e725b28968c650dbf306f43441892f09; 22 estimate

records 65a31d9f8c73459aa29c3bc4388ed22a156952a18d71388440bffb38f4c073c4; production manifest fb91f91676b84babbcb09ee6cd9129e58580249e21d3b223d71880170873bf4c; production completion marker ea9763c421161fe8044f6c489b09183f743f4cd8514ab9bab67424f88c1949ff; production-output authentication 21f8e4147b68ba8099b7d0c293f512ec51e269b40e4b816c2c9b81b826e2a865; failed audit attempt a54f2842774f38af0cab5a5ee6564b8ac295baf25549b7695b9f4d0500c45531; normalization receipt b66baa25585d2a7d0cb4a1ca3fdc002ed8bc7f3ca9f92b6c48475e64da786b5a; recovery replay receipt 497970f5243313a6cd945b69ce9c7eaba7449ceb33ef6d0dfd0d8873811ec5cc; recovery Python verification receipt 151f88a11c42d36bc9e64c97d98d83afe087e579bfbde4e00762ed13ae73ab11; recovery runner receipt 6ef9d18b51125253b57d59d32c1694a33bda428f54fc83351da0a7d62fee0d5e; acceptance record 2ac18b3f6c22bc6f11ebbf21ffd8614dbcf24f8f3d028c169e92b2d046b02d5d (accepted at 20:54:51 UTC on 2026-10-02). The frozen Study 003 specification, revision 1, has SHA-256 0ff662dc31807b0b2cdcae371d884473ed398e2583272d94331bac8c8f9ff0b0, and its design-GO record has SHA-256 e5dc485dd858f18eb9652cfe6b63f142e2d09fb6b069d640438cd8dbe63a786d; both are bound in the frozen decision record. The post-audit source patch, recorded at 20:56:31 UTC on 2026-10-02 after acceptance, changes one line of the producer's audit sink so that the three non-chunk audit headers carry 0xFFFFFFFF. It was neither compiled nor executed, it changes no scientific operator, record or output, and no scientific rerun was warranted.

BIT002 availability: the accepted release includes all 45 production data files, the 107-file producer/analyzer and 26-file independent-auditor sources, 33 keys and public fixtures, three accepted Linux binaries, the 67-file runtime package, build identities, exact results and audit/gate receipts. Its result hash is 190dd7f33dc48ff5a2b8f61c4650257cda435b026eb16ea8ec34a763152d9cb2. The companion evidence directory supplies the full accepted addendum, all six exact results, derived intervals, complete eligibility histograms and original descriptives. The prior G9 acceptance covers the BIT002 addendum. A separate nonauthor content review accepted the full integrated manuscript; both remain internal computational reviews, not journal peer review. Existing release manifests remain unchanged.

8. AI assistance

AI systems assisted with design criticism, implementation, code review, analysis checking, literature organization and drafting, under human direction. The author directed the work, made the scientific decisions and is responsible for the content. Internal reviews performed by AI models, including the non-implementing AI review of the frozen specification, are not external peer review. The integration of Study 002 into version 1.1.0 and of Study 003 into version 1.2.0 was also drafted with AI assistance. The integration of the subsequent BIT002 immediate-intervention experiment into revision 1.3.0 was drafted and checked with AI assistance, using the previously accepted results without new scientific runs.

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